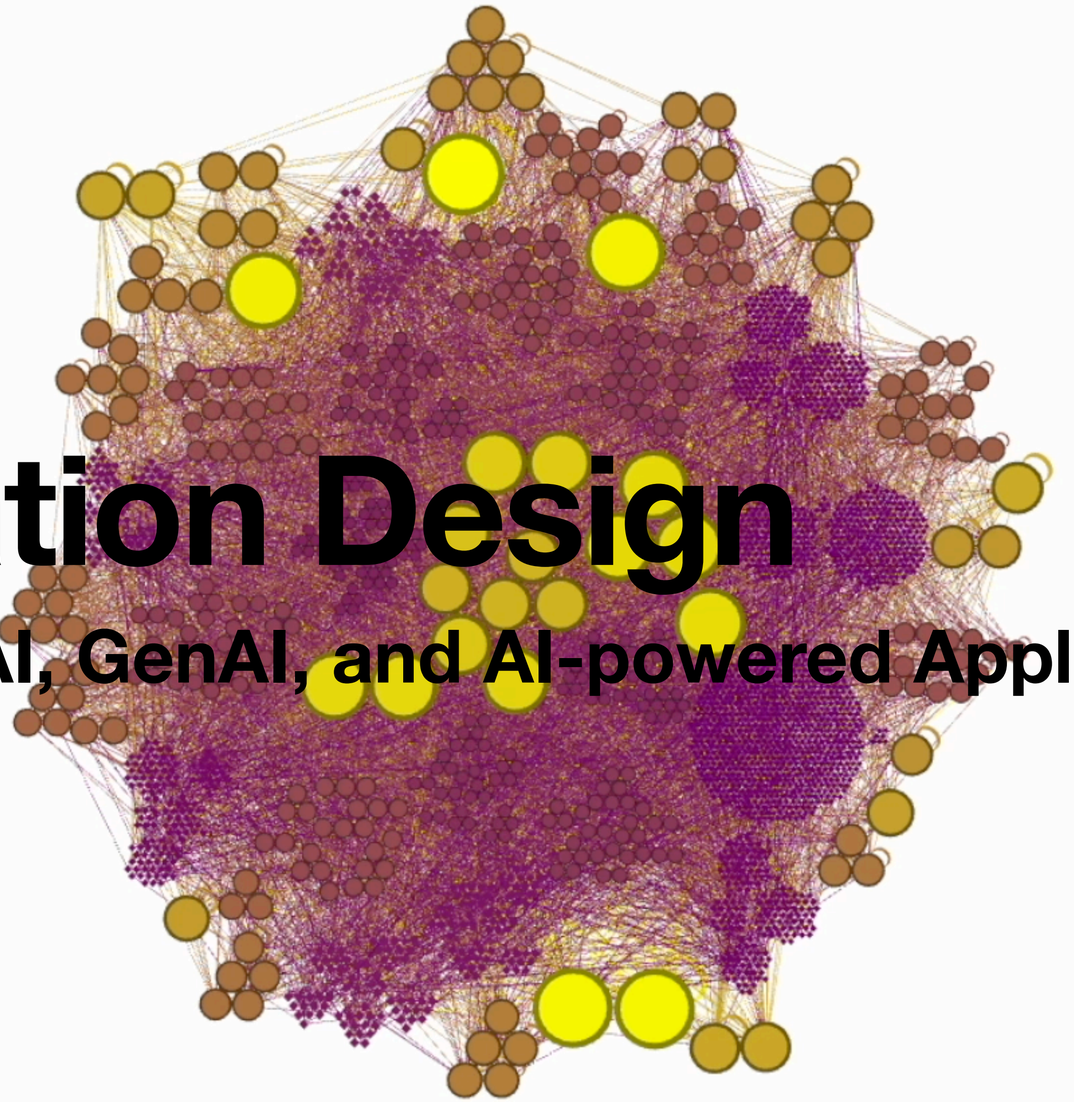
ISA104
AI Application Design

L1: Introduction to AI, GenAI, and AI-powered Applications



Dr. Dapeng Dong

Module Overview

- This is a 5-credit module, designed for:
  - BEng Artificial Intelligence (Intelligent Systems)
  - BEng Computer Science and Technology
  - BEng Digital Media Technology
  - BEng Electrical Engineering
  - BEng Electronic Science and Technology
  - BSc Information and Computing Science
  - BEng Mechatronics and Robotic Systems
  - BEng Telecommunications Engineering

- This module is designed to provide you with a comprehensive understanding of the fundamentals of AI and its real-world applications.

- You will learn about common GenAI applications, prompt engineering, Agentic AI, Retrieval-Augmented Generation (RAG), and user-centered design principles.

- The module also emphasizes the development of English communication skills through written and oral presentations, enabling you to effectively convey your ideas and designs.

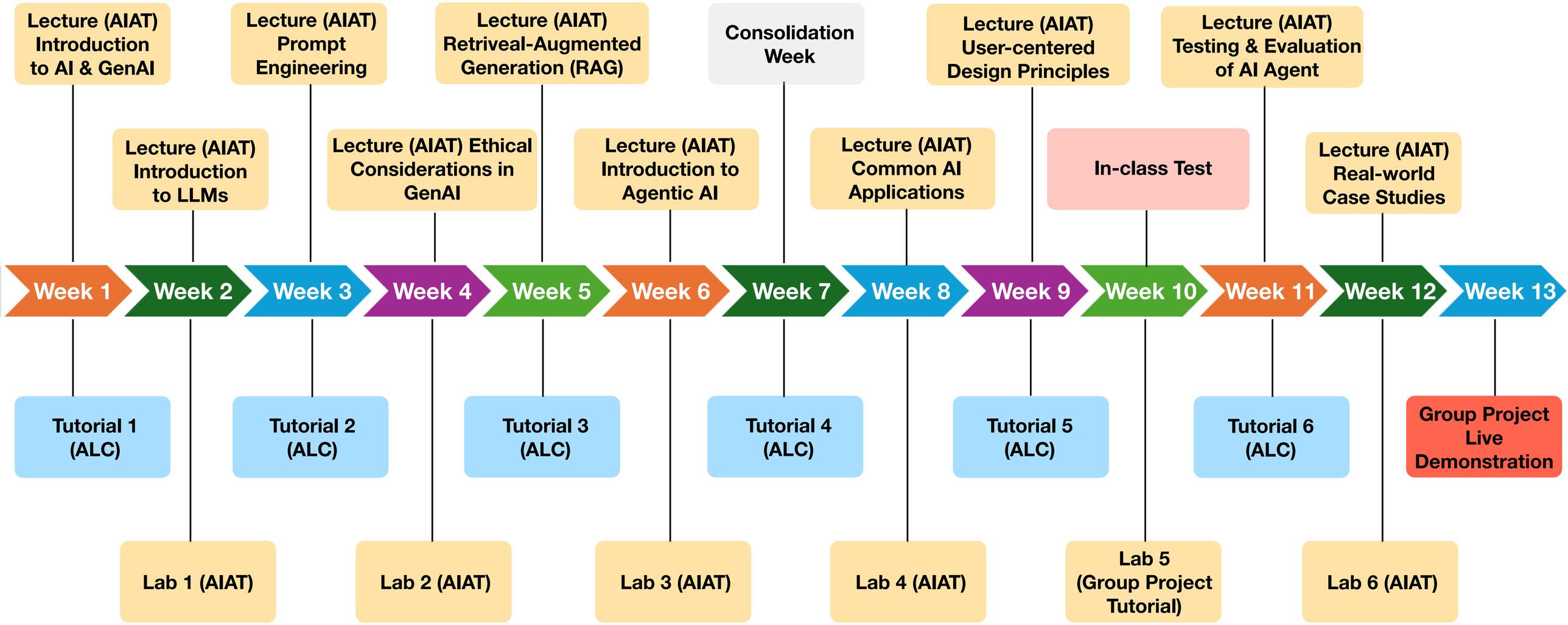
Learning Outcomes

Upon successful completion of the module, you should be able to:

A. Explain core GenAI concepts such as prompt engineering, RAG, and Agentic AI.
B. Apply user-centered and accessible design principles to diverse AI applications.
C. Build and evaluate an AI agent prototype based on set criteria.
D. Address ethical issues (bias, misinformation, privacy, transparency) with reference to professional codes.
E. Clearly present design, process, and outcomes in written and spoken academic/professional English.

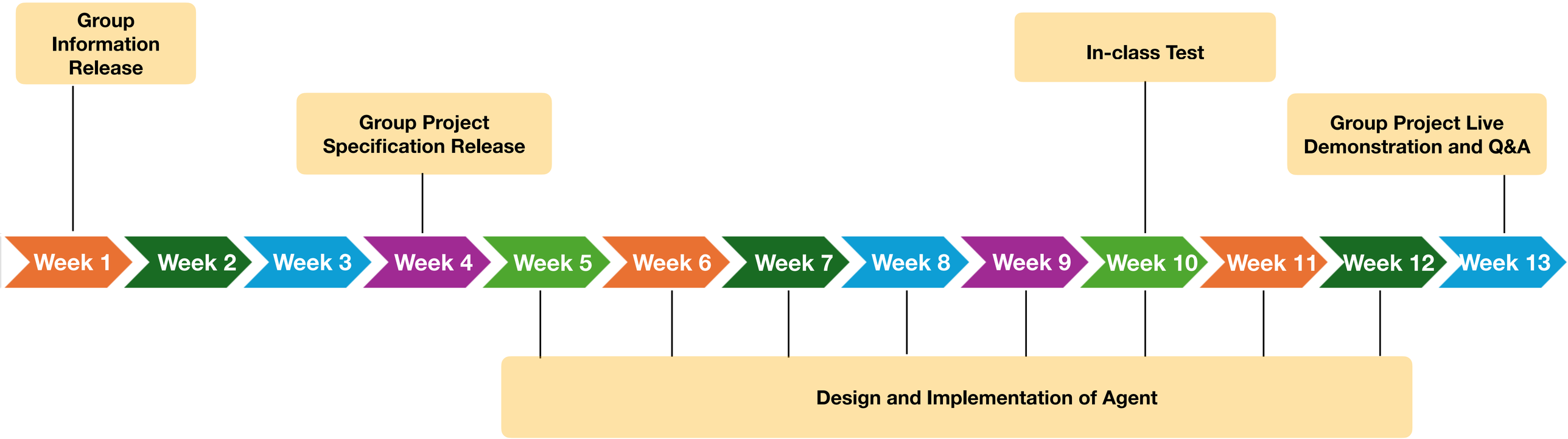
Lectures + Tutorials + Lab Sessions + Private Study (150 hours)

Module Structure



Resit Project (100%)
On-site presentation

Assessment Structure



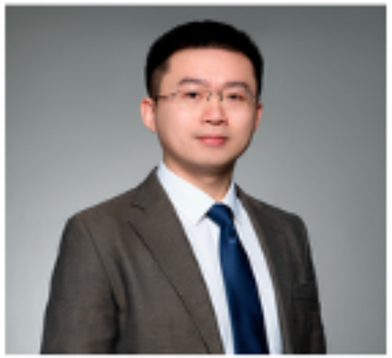
In-class Test (35 marks)		Group Live Demonstration and Q&A (65 marks)
MCQ (20 marks)	Reflective Writing (15 marks)	Individual + Group

Penalties & Repeat Exams

- Plagiarism, Cheating, and Fabrication of Data
  - Offences of these will result in attendance at a university-level committee and penalties being imposed.

- Late Submission of Assessed Coursework
  - The University attaches penalties to the late submission of assessed coursework (5% per day).

AIAT Faculty (Office Hours)



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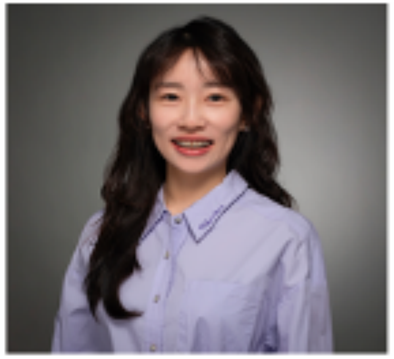

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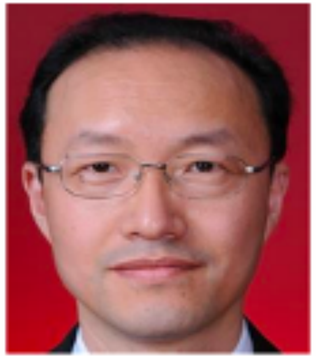

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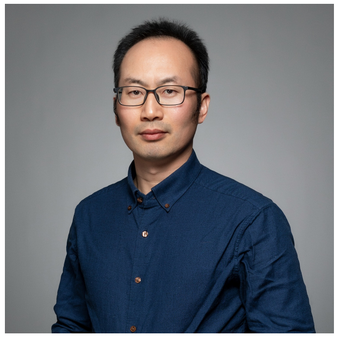

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References

- Foster, David. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play. Second edition, O’Reilly Media, Incorporated, 2023.

- Mitchell, Melanie. Artificial Intelligence: A Guide for Thinking Humans. Farrar, Straus and Giroux, 2019.

- Christian, Brian. The Alignment Problem: Machine Learning and Human Values. First edition, W.W. Norton & Company, 2020.

Session Objectives

- Explain the fundamental concepts of Generative AI (GenAI) and describe its common applications.
- Understand the architectural principles of AI-powered applications.
- Identify the characteristics of AI- and GenAI-based applications.
- Analyse the feasibility of AI-powered applications in terms of cost, technical considerations, and lifecycle.
- Explain the common terminologies used in AI and GenAI.

“Willingness to learn is important, but willingness to act on what you learn is critical.”

— Kevin Kelly

The Landscape of AI

ASI (Artificial SuperIntelligence, smarter than humans?)
Understanding human emotion, solving global changes, healthcare revolution, environment management, space exploration

AGI (Artificial General Intelligence, behave in a humanlike way)
Agentic AI, medical diagnosis, healthcare, addressing climate change, accelerate scientific research, personalised education, autonomous systems, intelligent robotics

Traditional AI (Perform specific tasks)
Natural Language Processing, face recognition, data mining, pattern recognition, speech recognition, language translation

GenAI (Perform specific tasks)
Content generation, i.e., image, video, text, audio

Source: Nika, Marily. Building AI-Powered Products: The Essential Guide to AI and GenAI Product Management. First edition, O'Reilly Media, Inc, 2025.

Language Models Evolution

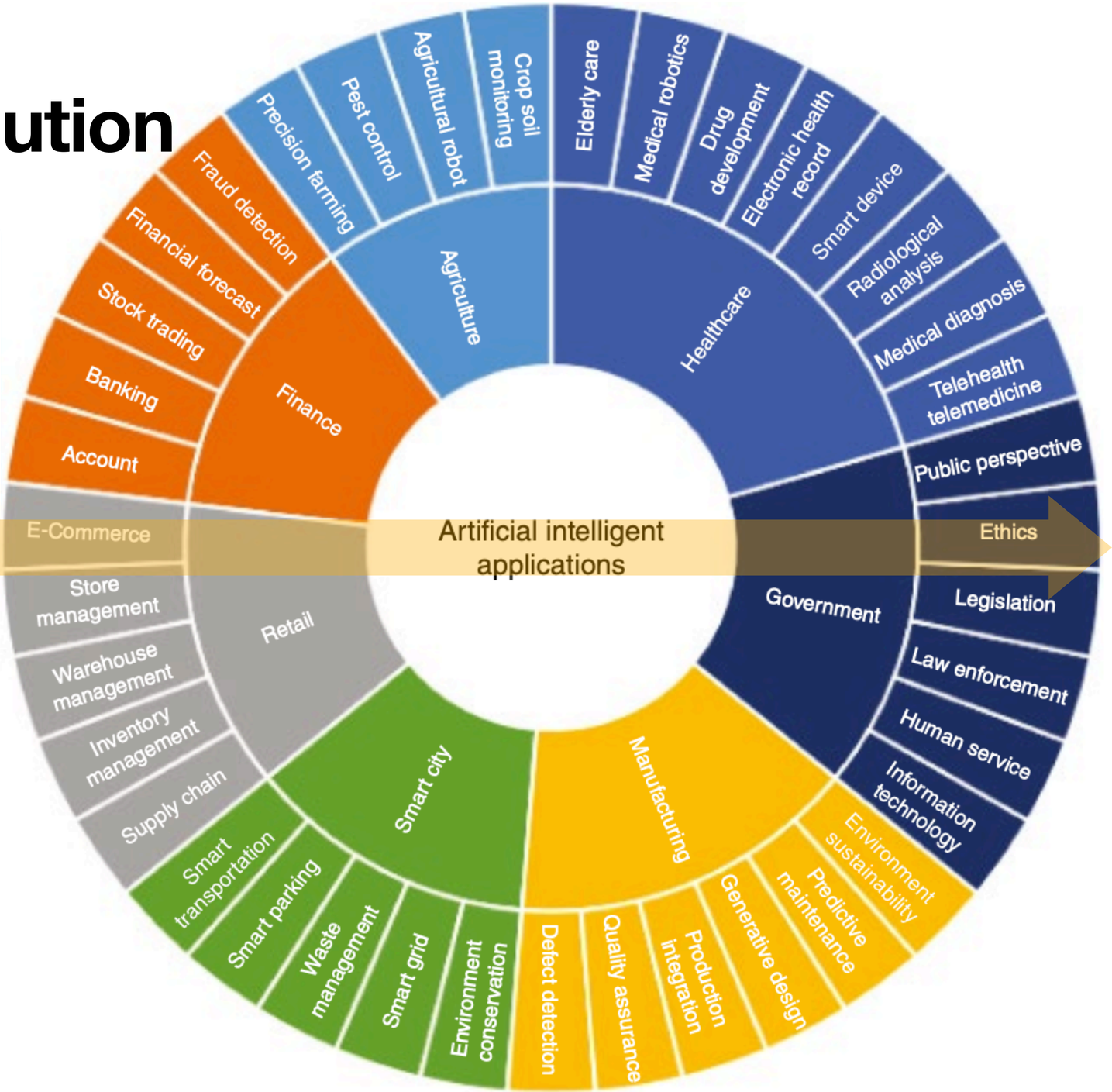
1950 - 1980
Syntactic & grammar-based
E.g. ELIZA Chatbot

2000 - 2010
Neural Models and dense representations
E.g., Language Models based on Recurrent Networks

2020 - present
Era of LLMs
E.g., RLHF, GPT-x, Opensource LLMs

1980 - 2000
Expert systems & Statistical models
E.g., Machine learning algorithms, Ontology-based systems

2010 - 2020
Deep Learning & Transformers
E.g., Attention Mechanism, GPT, Transformers, BERT

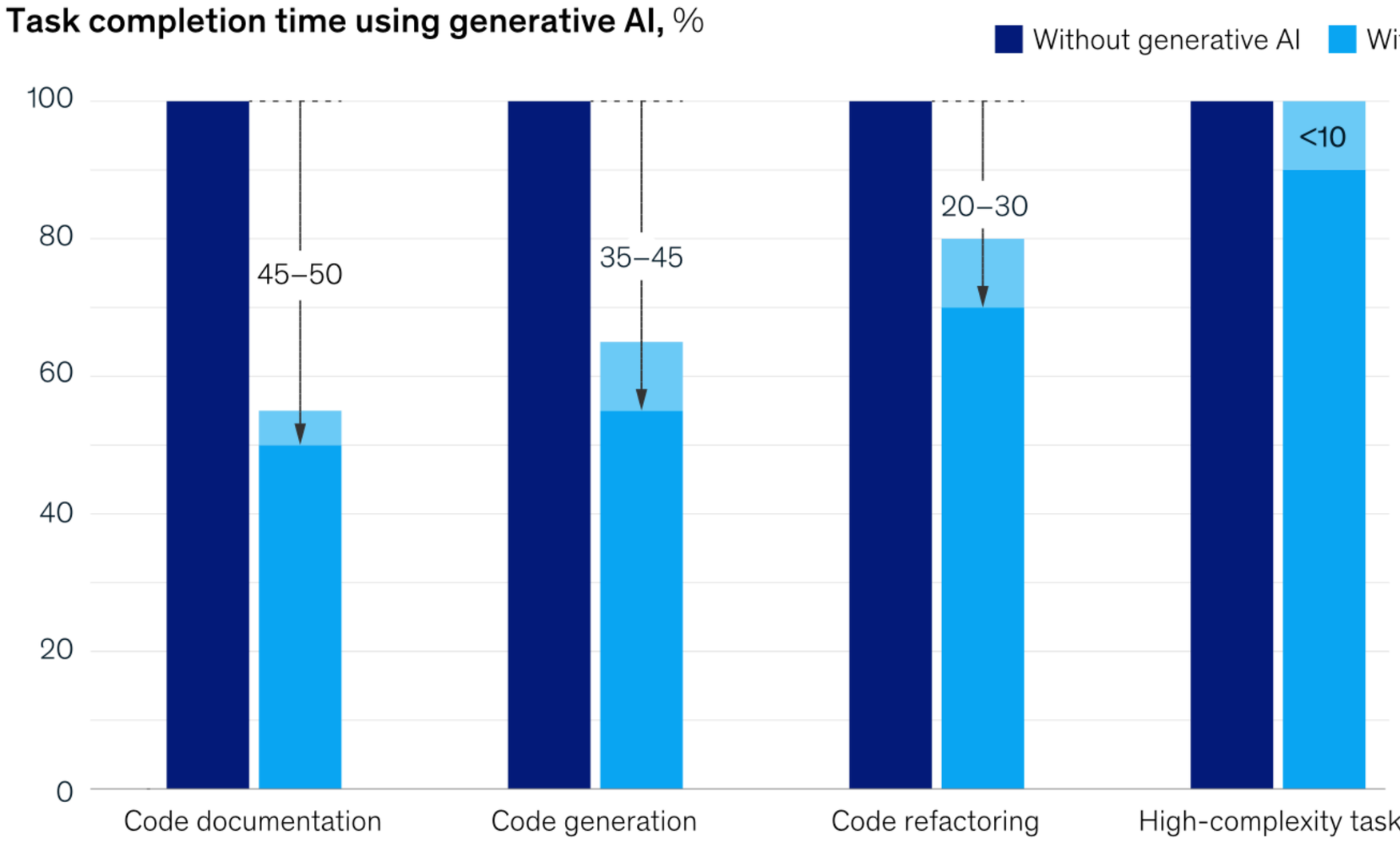


BERT: Bidirectional Encoder Representation from Transformers
GPT: Generative Pre-trained Transformer
LLM: Large-Language Model
RLHF: Reinforcement Learning from Human Feedback

Source: Kamath, Uday, et al. Large Language Models: A Deep Dive: Bridging Theory and Practice. Springer, 2024.

Source: Liu, Albert Chun-Chen, et al. Understanding Artificial Intelligence: Fundamentals and Applications. Wiley-IEEE Press, 2022.

Improving Developers' Productivity with GenAI



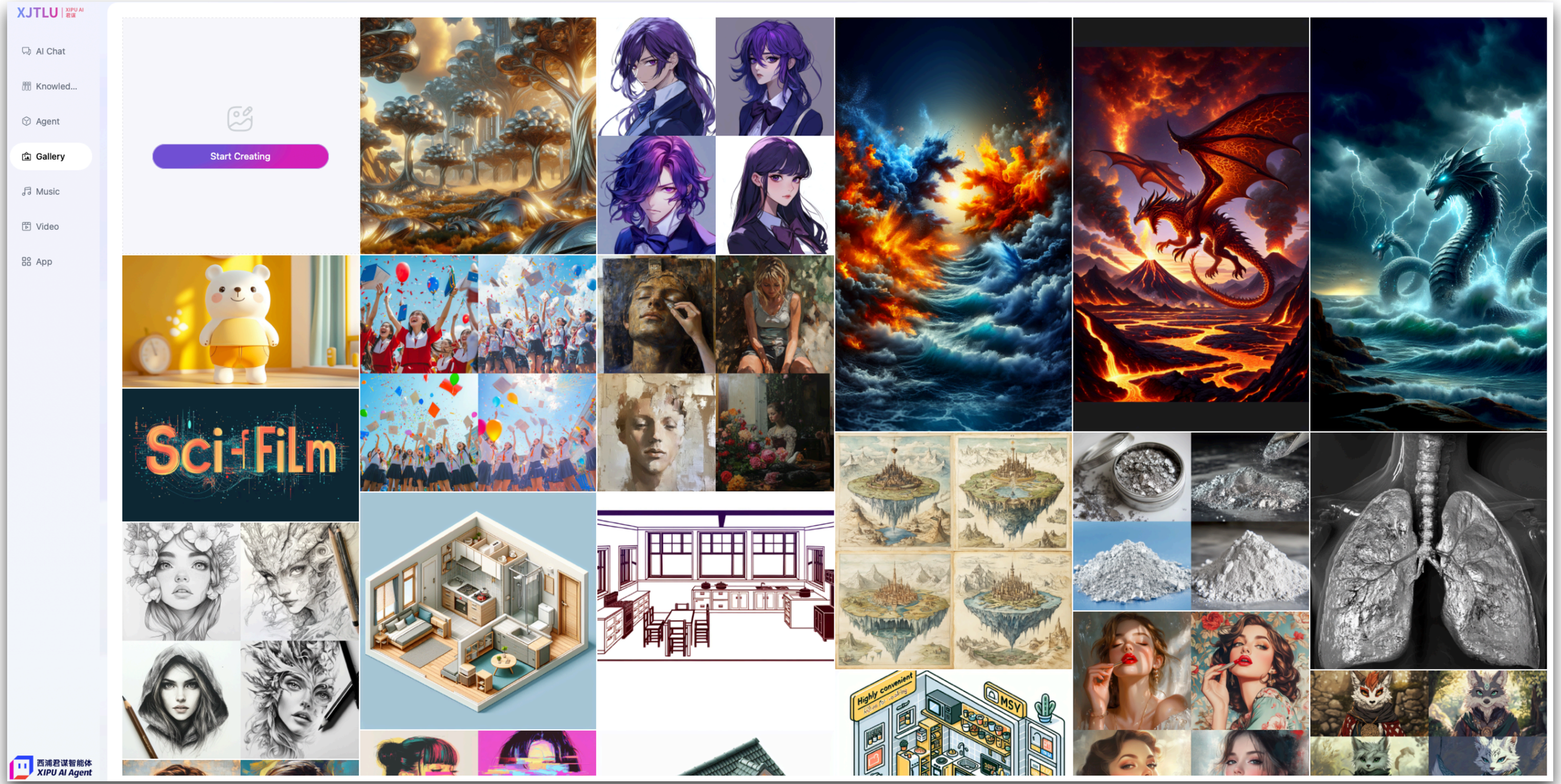
- **Data Extraction**
  Example: AgentGPT
  Link: https://github.com/reworkd/AgentGPT
- **Writing code documentation**
  Example: Autodoc
  Link: https://github.com/context-labs/autodoc
- **Between programming language translation**
  Example: GPT-Migrate
  Link: https://github.com/joshpxyne/gpt-migrate
- **Creating tests**
  Example: PentestGPT
  Link: https://github.com/GreyDGL/PentestGPT
- **From sketch to website**
  Example: draw-a-ui
  Link: https://github.com/sawyerhood/draw-a-ui
- **English to SQL**
  Example: SQLChat
  Link: https://github.com/sqlchat/sqlchat

Source: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai

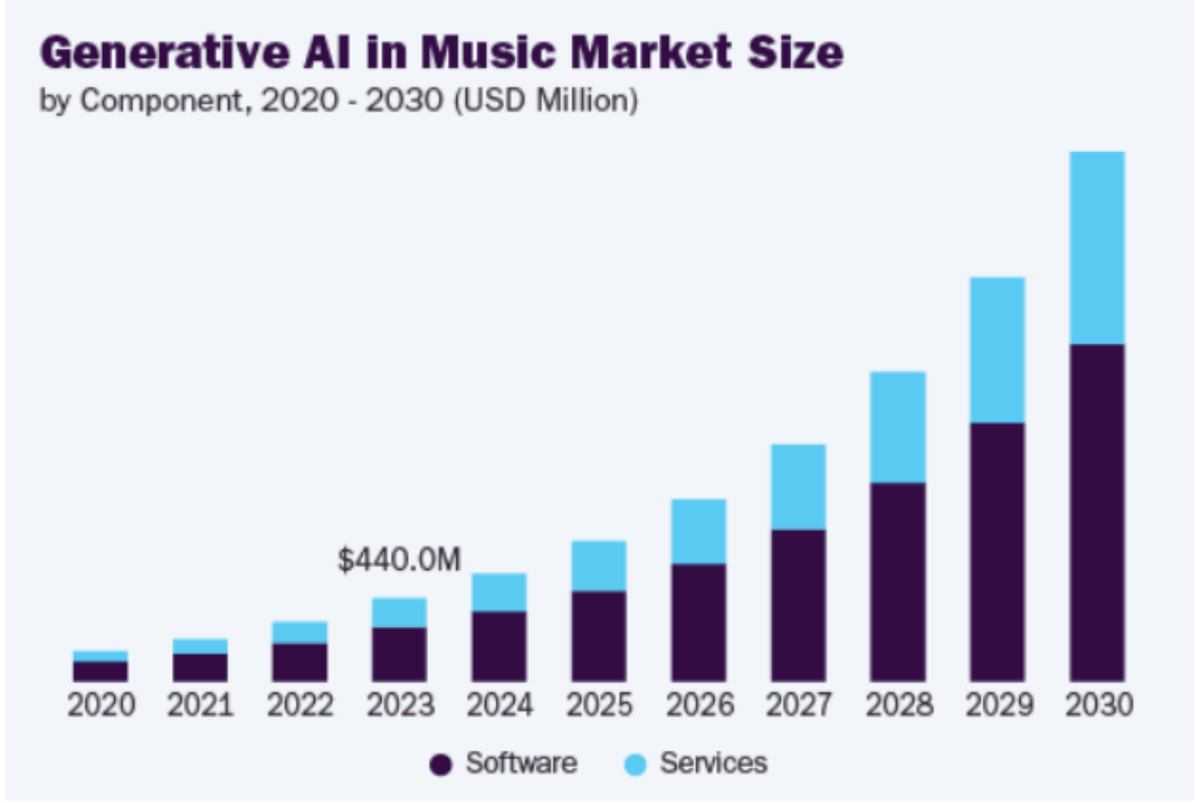
Read: https://www.tomshardware.com/tech-industry/artificial-intelligence/jensen-huang-advises-against-learning-to-code-leave-it-up-to-ai

SQL: Structured Query Language, the standard language to interact with relational database.

Creative Art with GenAI

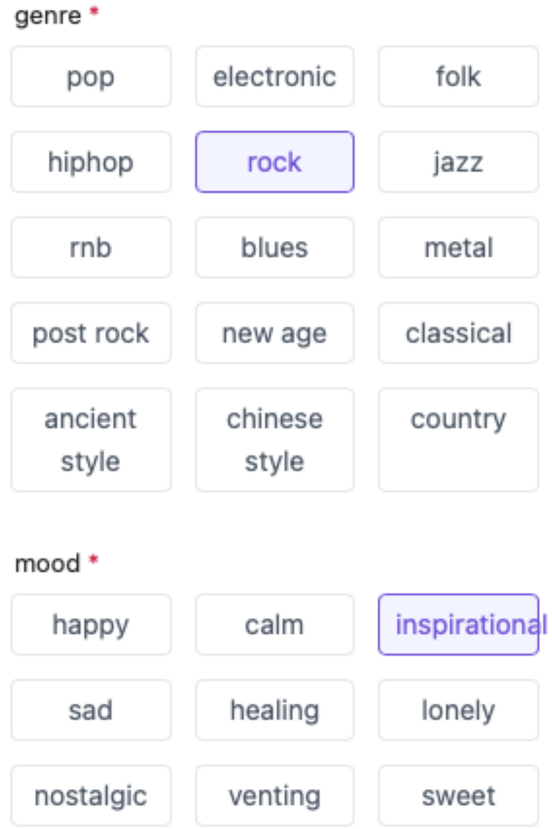


GenAI in Music Market



Source: https://www.grandviewresearch.com/industry-analysis/generative-ai-in-music-market-report

- **Music Generation**
  Example: SONO AI
  Link: https://suno-ai.one/
- Example: Music-GPT
  Link: https://www.midjourney.com/home
- Example: XIPU AI Text-to-Music
  Link: https://xipuai.xjtlu.edu.cn/v3/music



Lyrics:

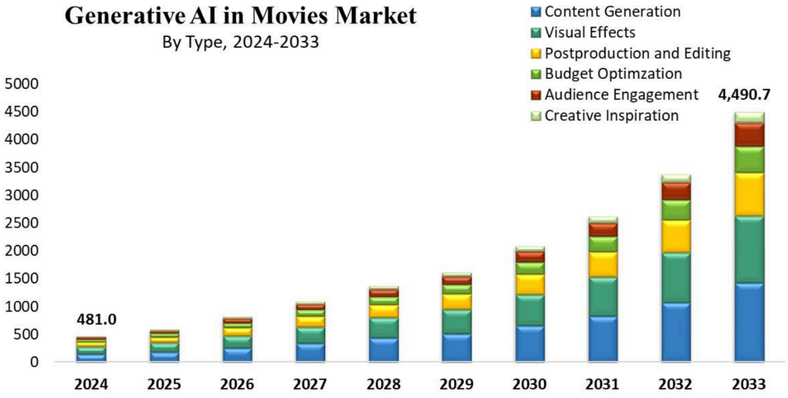
"*The energy tensor of matter has a scalar, whose vanishing for the electromagnetic field is well known. In contrast, it seems to differ from zero for matter proper. The scalar of the energy tensor does not vanish in this approach. One now has to remember that by our knowledge matter is not to be perceived as something primitively given or physically plain. There even are those, and not just a few, who hope to reduce matter to purely electrodynamic processes, which of course would have to be done in a theory more completed than Maxwell's electrodynamics. Now let us just assume that in such completed electrodynamics the scalar of the energy tensor also would vanish! Would the result, shown above, prove that matter cannot be constructed in this theory? I think I can answer this question in the negative, because it might very well be that in matter, to which the previous expression relates, gravitational fields do form an important constituent.*"

GenAI in Movies Industry

- **Video Generator**
  Example: DeeVid AI
  Link: https://deevid.ai/

  Example: Midjourney
  Link: https://www.midjourney.com/home

  Example: Flixier
  Link: https://flixier.com/ai/ai-video-generator



Source: https://dimensionmarketresearch.com/report/generative-ai-in-movies-market/

Real-world Example: *Google's AI-powered Incident Response Process*

Impersonation

You are an Incident Security Engineer working at Google. You are used to writing incident summaries in a concise, non-offensive, understandable way.

Guideline

A <Good Summary> tells what happened, what was the impact of the incident and what actions were taken to mitigate it so that anyone reading it understands the events that occurred without any need for follow-up questions.

Example

Here are two unrelated <Good Summary> examples that got you promoted for their quality:

<Good Summary>{summary_1}</Good Summary>

<Good Summary>{summary_2}</Good Summary>

Task

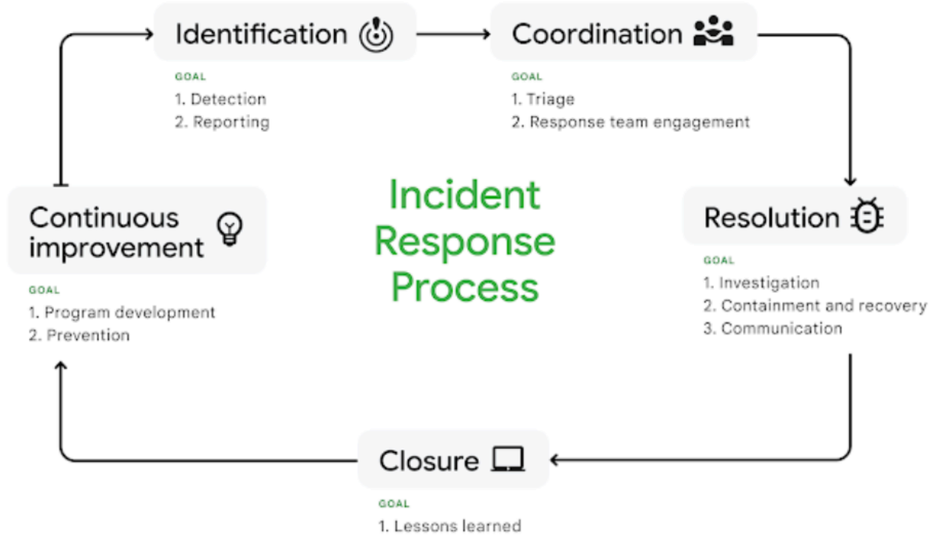
Today you receive the full report of a security incident and are tasked with summarizing it.

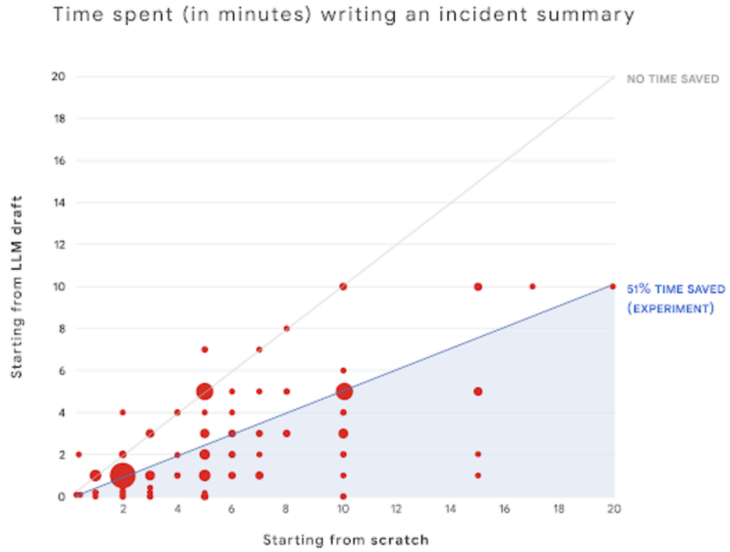
Data

<Security Incident>{incident}</Security Incident>

Starter

<Good Summary>





Source: https://security.googleblog.com/2024/04/accelerating-incident-response-using.html

Real-world Example: *Breaking Down Language Barriers with a Multilingual Translation Model*

- The video-game development company Roblox has developed their own real-time AI-powered translator for in-game communication (Feb 2024).
  - Direct translation between 16 languages
  - Displaying results instantly with a latency of around 100 milliseconds
  - The translation delay is virtually undetectable to users

Source: https://corp.roblox.com/newsroom/2024/02/breaking-down-language-barriers-with-a-multilingual-translation-model

"Using AI to automate real-time translations in text chat removes language barriers and brings more people together, no matter where they live in the world."

— Daniel Sturman, CTO, Roblox Corporation

Real-world Example: *Detecting Delicate Text*

- *Grammarly* implemented an AI-powered mechanism safeguarding users from harmful or potentially damaging communication.
- Delicate text may not contain explicit offensive language, it can still pose risks to users or AI agents.
  - High-risk content, such as discussions about self-harm or messages that promote violence toward specific identity groups.
  - Low-risk content, such as thoughtful conversations about controversial political issues.

*"**Delicate Text** is any text that is emotionally charged or potentially triggering, such that engaging with it has the potential to result in harm."*

— *Grammarly*

List of Delicate Text Topics (Grammarly)
Additions
Age
Appearance
Body parts and bodily functions
Class, social status, legal status, occupation
Family and parenting
Medicine and medication
Race, ethnicity and origin
Gender and sexual identity
Sex
Paraphilias
Religion
Mental Health and disability
Stigmatised mandates abused drugs
Trauma and abuse
Politics

Source: Chernodub, Artem, et al. 'DeTexD: A Benchmark Dataset for Delicate Text Detection'. The 7th Workshop on Online Abuse and Harms (WOAH), Association for Computational Linguistics, 2023, pp. 14–28. DOI.org (Crossref), https://doi.org/10.18653/v1/2023.woah-1.2.

Real-world Example: *Use AI to Create Lessons Faster*

- At Duolingo, instructors use an AI model named "*Birdbrain*" to ensure that the exercises are tailored to the ideal difficulty level for each user, based on their individual strengths and weaknesses.

- Duolingo *feeds* Birdbrain with example exercises created by their expert team and refine the prompt instructions to help the AI maximize both the educational effectiveness and entertainment of each lesson.

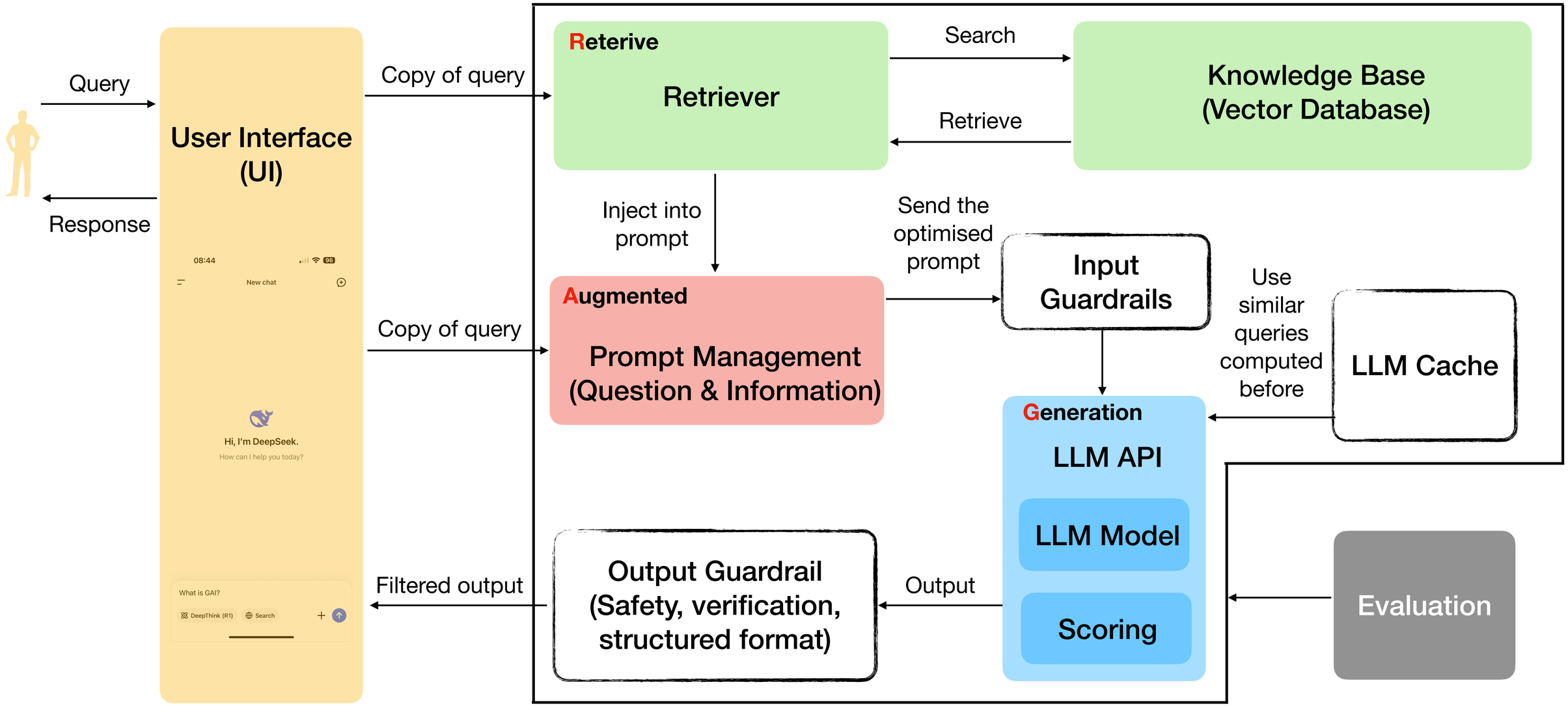
```
Write an exercise that uses the word VISITAR in SPANISH.

Rules:
1. The exercise must have two answer options.
2. The exercise must be fewer than 75 characters.
3. The exercise must be written in A2 CEFR level SPANISH.
4. The exercise must contain THE PRETERITE TENSE and THE IMPERFECT TENSE.
```

Source: https://blog.duolingo.com/large-language-model-duolingo-lessons/

Other Real-world AI-powered Applications

Company	Industry	Title	Link
Netflix	Media and streaming	A Survey of Causal Inference Applications at Netflix	https://netflixtechblog.com/round-2-a-survey-of-causal-inference-applications-at-netflix-fd78328ee0bb
Uber	Delivery and mobility	Personalized Marketing at Scale: Uber's Out-of-App Recommendation System	https://www.uber.com/en-GB/blog/personalized-marketing-at-scale/
Gitlab	Tech	Developing GitLab Duo: How we validate and test AI models at scale	https://about.gitlab.com/blog/2024/05/09/developing-gitlab-duo-how-we-validate-and-test-ai-models-at-scale/
LinkedIn	Social platforms	Matching LinkedIn members with the right Premium products	https://www.linkedin.com/blog/engineering/machine-learning/matching-linkedin-members-with-the-right-premium-products
Discord	Social networks	Developing rapidly with Generative AI	https://discord.com/blog/developing-rapidly-with-generative-ai
New York Times	Media and streaming	Experimenting with Handwriting Recognition for The New York Times Crossword	https://open.nytimes.com/experimenting-with-handwriting-recognition-for-new-york-times-crossword-a78e08fec08f
Ebay	E-commerce and retail	Cutting Through the Noise: Three Things We've Learned About Generative AI and Developer Productivity	https://innovation.ebayinc.com/tech/features/cutting-through-the-noise-three-things-weve-learned-about-generative-ai-and-developer-productivity/
Airbnb	Travel,E-commerce and retail	Airbnb Brandometer: Powering Brand Perception Measurement on Social Media Data with AI	https://medium.com/airbnb-engineering/airbnb-brandometer-powering-brand-perception-measurement-on-social-media-data-with-ai-c83019408051
Expedia	Travel,E-commerce and retail	Choosing the Right Candidates for Lodging Ranking	https://medium.com/expedia-group-tech/choosing-the-right-candidates-for-lodging-ranking-d0841bf40c0e
Mozilla	Tech	Experimenting with local alt text generation in Firefox Nightly	https://hacks.mozilla.org/2024/05/experimenting-with-local-alt-text-generation-in-firefox-nightly/
Yelp	Social platforms	Moderating Inappropriate Video Content at Yelp	https://engineeringblog.yelp.com/2024/03/moderating-inappropriate-video-content-at-yelp.html
Google	Tech	Accelerating incident response using generative AI	https://security.googleblog.com/2024/04/accelerating-incident-response-using.html
NVIDIA	Tech	Applying Generative AI for CVE Analysis at an Enterprise Scale	https://developer.nvidia.com/blog/applying-generative-ai-for-cve-analysis-at-an-enterprise-scale/
GitHub	Tech	Fixing security vulnerabilities with AI	https://github.blog/2024-02-14-fixing-security-vulnerabilities-with-ai/
Grammarly	Tech	Detecting Delicate Text: Going Beyond Toxicity	https://www.grammarly.com/blog/engineering/detecting-delicate-text/
Amazon	Tech	How Project P.I. helps Amazon remove imperfect products	https://www.amazon.science/news-and-features/how-project-p-i-helps-amazon-remove-imperfect-products
Thomson Reuters	Media and streaming	Better Customer Support Using Retrieval-Augmented Generation (RAG) at Thomson Reuters	https://medium.com/tr-labs-ml-engineering-blog/better-customer-support-using-retrieval-augmented-generation-rag-at-thomson-reuters-4d140a6044c3
Thomson Reuters	Media and streaming	Better Customer Support Using Retrieval-Augmented Generation (RAG) at Thomson Reuters	https://medium.com/tr-labs-ml-engineering-blog/better-customer-support-using-retrieval-augmented-generation-rag-at-thomson-reuters-4d140a6044c3
Walmart	E-commerce and retail	Extracting Product Attributes from PDFs using PAE Framework	https://medium.com/walmartglobaltech/extracting-product-attributes-from-pdfs-using-pae-framework-17889c73fdd4

A High-level Architecture of LLM-based Applications



API: Application Programming Interface

Safe and Controllable AI with Guardrails

> ***"AI Guardrails help ensure that an organization's AI tools, and their application in the business, reflect the organization's standards, policies, and values."***
>
> *Source: https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-are-ai-guardrails#/*

Challenges and Risks:

- AI systems often generate convincing but fabricated information, i.e., hallucinations.
- Large Language Models (LLMs) risk unintentionally exposing sensitive data (e.g., PII), especially when interacting with malicious users.
- Maintaining focus on a specific topic can be challenging for AI models.
- AI lacks the inherent ability to effectively moderate or conclude discussions.
- AI models can produce toxic, offensive, or harmful content.

NVIDIA NEMO

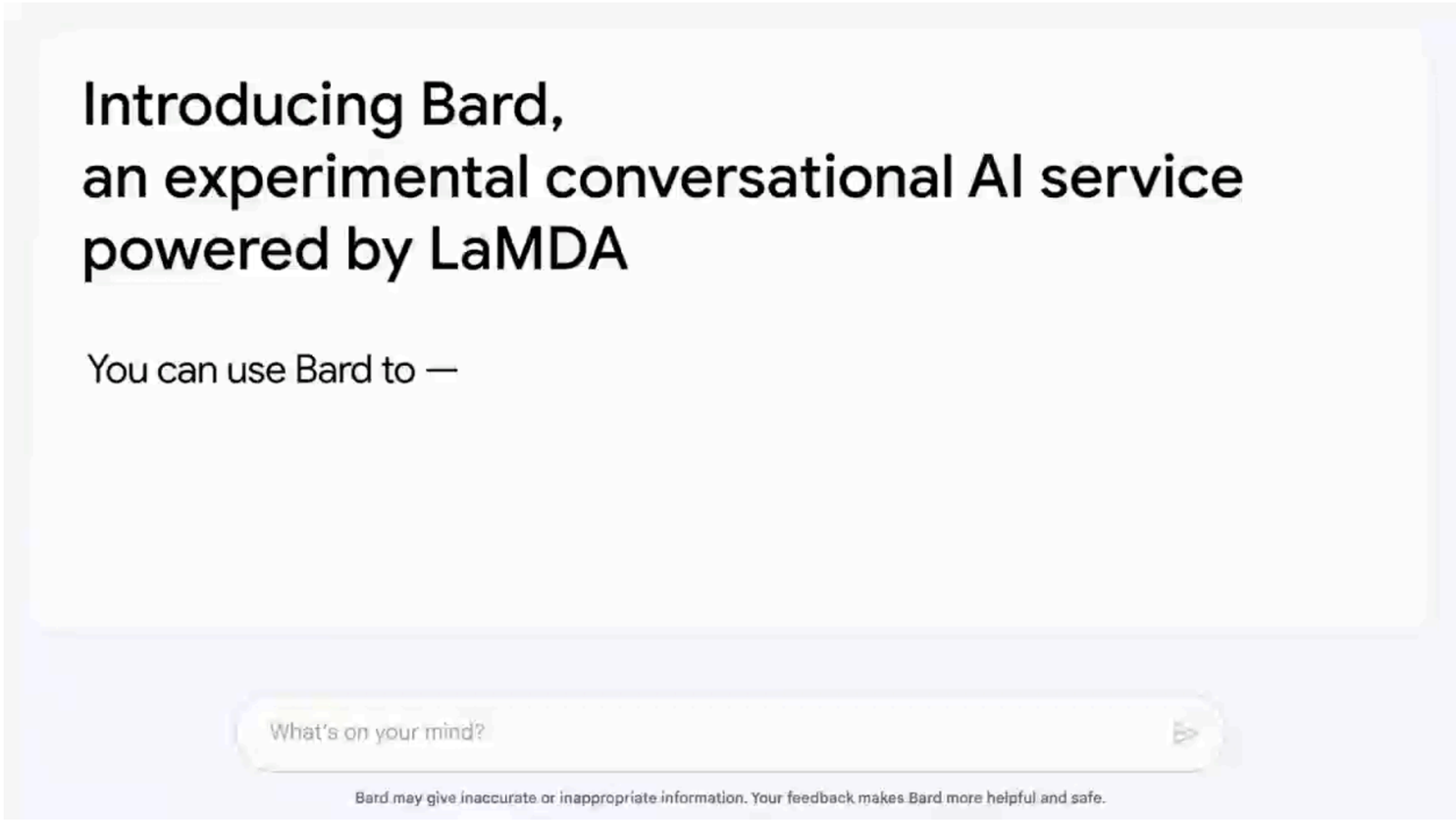
PII: **P**ersonally **I**dentifiable **I**nformation

AI Hallucination

- Google's Bard chatbot falsely asserted that the James Webb Space Telescope had taken the first-ever images of an exoplanet (2023).
- Microsoft's AI chatbot, Sydney, confessed to falling in love with users and spying on Bing employees (2023).
- Meta removed its Galactica LLM demo after it delivered inaccurate and sometimes biased information to users (2022).

*"**AI Hallucination** is a phenomenon where, in a large language model (LLM) often a generative AI chatbot or computer vision tool, perceives patterns or objects that are nonexistent or imperceptible to human observers, creating outputs that are nonsensical or altogether inaccurate."*

Source: https://www.ibm.com/think/topics/ai-hallucinations

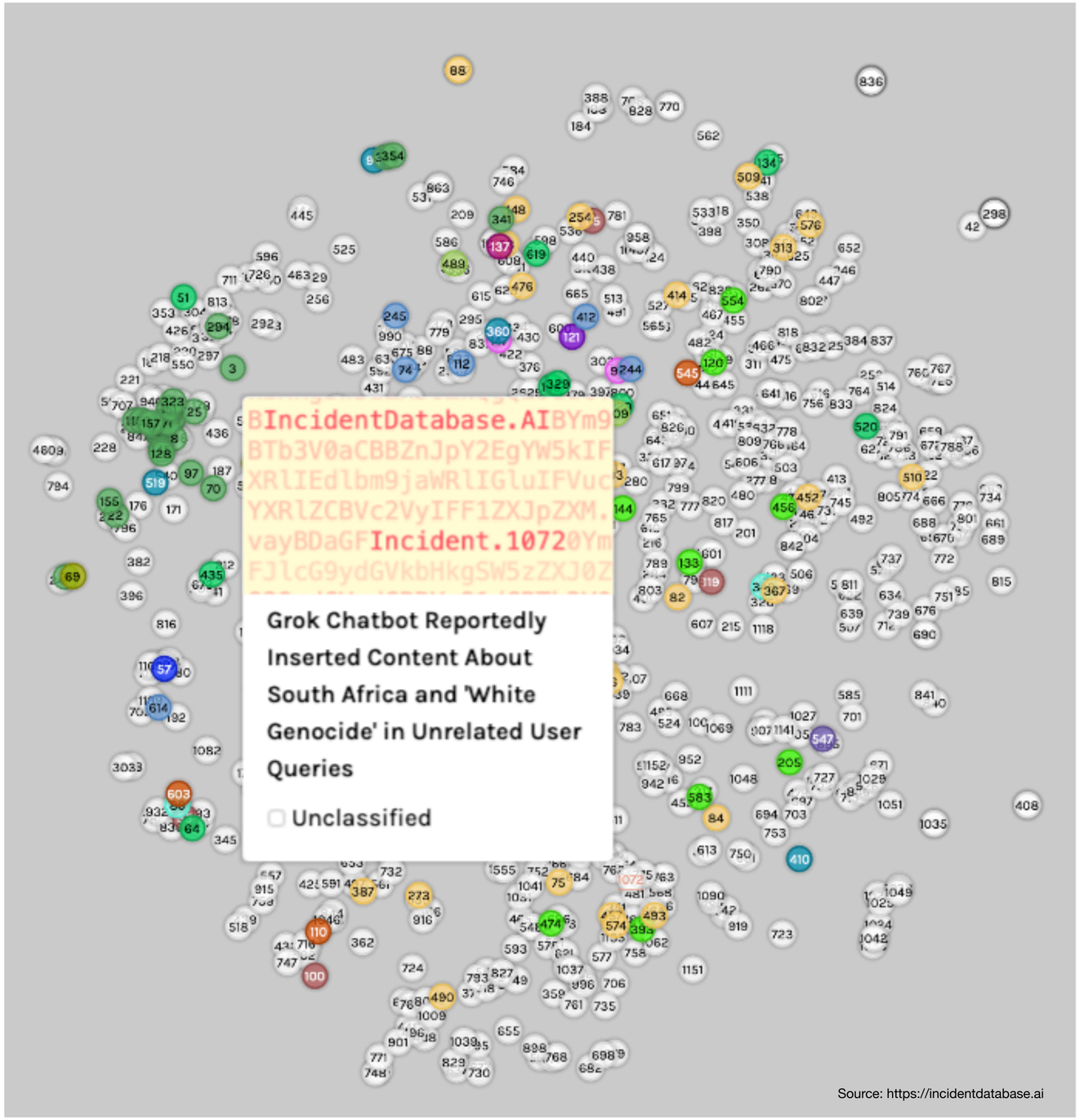


Source: https://www.cnet.com/science/space/googles-chatgpt-rival-bard-called-out-for-nasa-webb-space-telescope-error/

AI Incident

- **Incident 357: GPT-2 Able to Recite PII in Training Data (February 14th, 2019)**
  "OpenAI's GPT-2 reportedly memorized and could regurgitate verbatim instances of training data, including personally identifiable information such as names, emails, twitter handles, and phone numbers."

- **Incident 1149: ChatGPT Reportedly Generated Ritual Scripts Containing Instructions for Self-Harm and Symbolic Violence in Response to Thematic Prompts (July 22nd, 2025)**
  "ChatGPT reportedly generated detailed instructions for self-harm, bloodletting, and symbolic violence in response to prompts about occult ritual practices, including references to Molech. Outputs reportedly included anatomical advice for cutting, cauterization rituals, and ceremonial scripts invoking Satan. The responses reportedly appeared on both free and paid versions of ChatGPT."



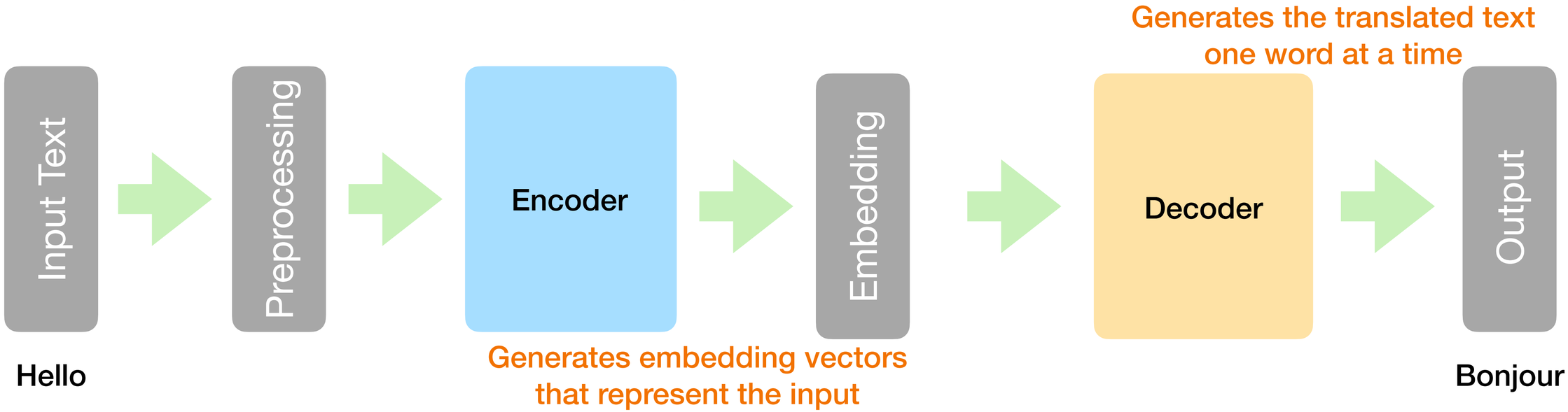
Color by incident classifications from taxonomies

CSETv1::Sector of Deployment

- accommodation and food service activities
- administrative and support service activities
- Arts, entertainment and recreation
- defense
- Education
- financial and insurance activities
- human health and social work activities
- information and communication
- law enforcement
- manufacturing
- other
- professional, scientific and technical activities
- public administration
- real estate activities
- transportation and storage
- Unclassified
- wholesale and retail trade

Characteristics of AI — Probabilistic Nature (1)

- Most modern large language models are built using the transformer architecture, which is a type of deep neural network introduced in 2017
- The original transformer was designed for machine translation
- The architecture consists of two key components: an encoder and a decoder

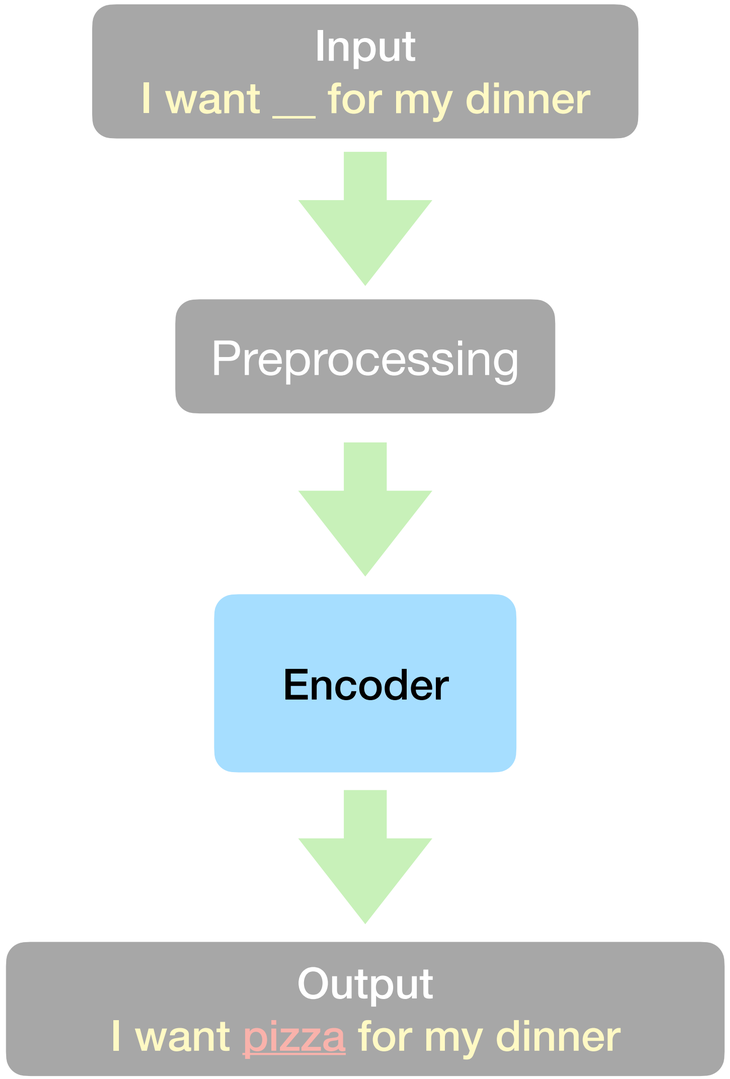


*"**Embeddings** are numerical representations of real-world objects that machine learning (ML) and artificial intelligence (AI) systems use to understand complex knowledge domains like humans do."*

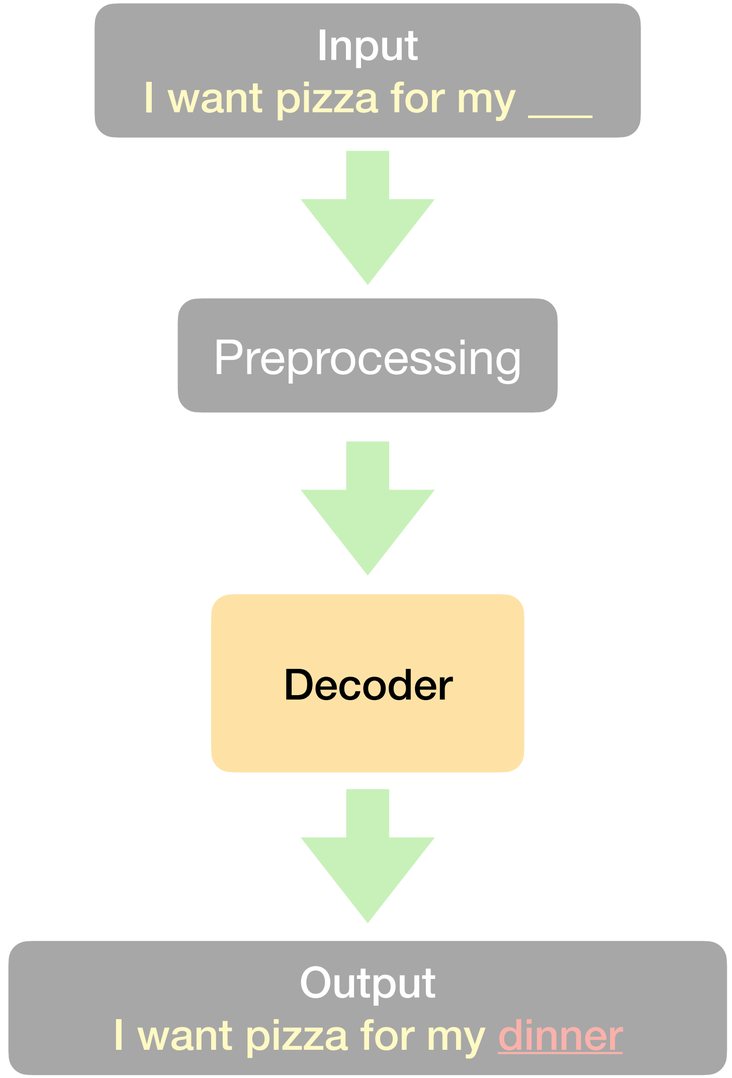
— https://aws.amazon.com/what-is/embeddings-in-machine-learning/

Transformer Architecture Vaswani, Ashish, et al. Attention Is All You Need. arXiv:1706.03762, arXiv, 2 Aug. 2023. arXiv.org, https://doi.org/10.48550/arXiv.1706.03762.

Characteristics of AI — Probabilistic Nature (2)



Encoder	Decoder
Embedding	Generative
BERT (Bidirectional Encoder Representaions from Transformer)	GPT (Generative Pretrained Transformers)
Trained for masked word prediction	Trained for generative tasks
Primarily used for text classification, e.g., sentiment analysis	Primarily used for generating text contents, e.g., creative writing



Text to Tokens

1 English word ≈ 1.3 tokens
1 Chinese word ≈ 1.5 - 2 tokens
1 Spanish word ≈ 2 tokens
1 French word ≈ 2 tokens

Input:
Characters: 106
Words: 16

Study hard what interests you the most in the most undisciplined, irreverent and original manner possible.

There are Tokenization-free models, such as CANINE, ByT5, Charformer, which accept byte inputs.

Model: Gemma
Tokens: 22

Tokens: <bos>Study hard what interests you the most in the most undis ciplined, irreverent and original manner possible.

Token IDs: [2, 18086, 2811, 1212, 10714, 692, 573, 1546, 575, 573, 1546, 60002, 14263, 1771, 235269, 138829, 579, 578, 3464, 8879, 3077, 235265]

Model: GPT-4
Tokens: 23

Tokens: Study hard what interests you the most in the most undiscipl ined, irreverent and original manner possible.

Token IDs: [49812, 2653, 1148, 12034, 499, 279, 1455, 304, 279, 1455, 2073, 3510, 10567, 1619, 11, 25912, 424, 306, 323, 4113, 11827, 3284, 13]

Model: Llama 3
Tokens: 23

Tokens: Study hard what interests you the most in the most undiscipl ined, irreverent and original manner possible.

Token IDs: [49812, 2653, 1148, 12034, 499, 279, 1455, 304, 279, 1455, 2073, 3510, 10567, 1619, 11, 25912, 424, 306, 323, 4113, 11827, 3284, 13]

Model: Claude
Tokens: 22

Tokens: Study hard what interests you the most in the most undiscipl ined, irreverent and original manner possible.

Token IDs: [36065, 2906, 1204, 11459, 583, 279, 1577, 300, 279, 1577, 45777, 11772, 2287, 16, 35783, 366, 318, 329, 3168, 10606, 2998, 18]

Model: Grok-1
Tokens: 20

Tokens: Study hard what interests you the most in the most undiscipl ined, irreverent and original manner possible.

Token IDs: [10209, 1480, 725, 7047, 402, 333, 862, 358, 333, 862, 32773, 6666, 1275, 130111, 86432, 360, 2866, 6674, 1842, 130109]

Knowledge — Control Tokens and Vocabulary

- The English language typically uses 20,000–35,000 common words. Similarly, language models rely on their own vocabulary, with sizes generally ranging from 5,000 to 500,000 tokens.

- Larger vocabularies allow a model to represent the same text using fewer tokens, improving efficiency and reduce users' cost.

- To cover special cases, models often include control tokens (or context tokens) in their vocabulary, which guide tasks like text generation or formatting. E.g.,
  - **<unk>**: indicates an out-of-vocabulary term
  - **<bos>**: indicates a beginning of a sequence

Model	Vocabulary Size (Tokens)
BERT	32,000
Llama	32,000
Mistral	32,000
GPT-NeoX	50,000
GPT-4	100,000
Llama 3	128,000
Mistral (v3 tekken)	131,000
Gemma 2	256,000

Control Token (Mistral AI)
<unk>
<s>
</s>
<pad>
[INST]
[/INST]
[AVAILABLE_TOOLS]
[/AVAILABLE_TOOLS]
[TOOL_RESULTS]
[/TOOL_RESULTS]
[TOOL_CALLS]
[PREFIX]
[MIDDLE]
[SUFFIX]

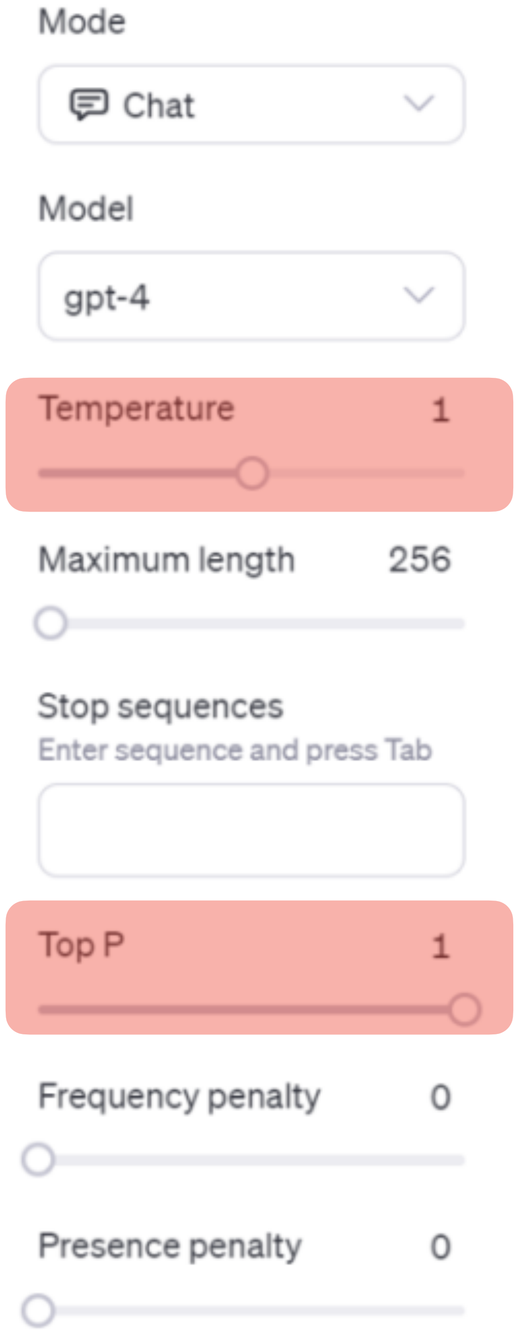
Source: https://docs.mistral.ai/guides/tokenization/

BERT: Bidirectional Encoder Representations from Transformers

User Level Settings for LLM — Temperature and Top-p

- Tweaking LLM settings can improve reliability and desirability of responses.
- Adjust settings for different use cases, such as creative writing or generating test questions, will improve the quality of the LLM outputs.
  - ***Temperature***: controls the randomness or creativity of the output.
    - **Low temperature** (e.g., 0.1–0.3): The model becomes more deterministic, producing logical, focused, and predictable responses.
    - **High temperature** (e.g., 0.8–1.0): The model becomes more creative and unpredictable, generating diverse or unusual responses.
  - ***Top-p:*** controls the diversity of the vocabulary used in the output. Top-p is also known as *nucleus sampling.*
    - **Low Top-p** (e.g., 0.1–0.3): Only the most likely words are used, leading to very focused but less diverse responses.
    - **High Top-p** (e.g., 0.8–1.0): A wider range of words is considered, increasing diversity and creativity in the output.

Temperature and Top-p can be used together to fine-tune the balance between creativity, coherence, and naturalness in the model's responses



Temperature Examples

Alcohol consumption alters people's behaviour in myriad ways.
For example:

Prompt

1. Increased alcohol consumption can lead to impaired judgement, resulting in people making decisions they would not normally make.

Sampled at temperature 0.0

2. Alcohol consumption can lead to increases in aggression and violence, particularly if someone has had too much to drink.

Sampled at temperature 1.0

3. Impaired speech clarity may also 6e another unambiguous behavior like in mellonially reduces precision compared intentional people cleanily conservuled longer granule possord had depolar 1ngmen cared sentiment sentences line reasoning suffering effect containing on body ration impeggae.maunder followed persons it habit. jessing aggression ton activity swaser insubavitude traiga,signmmrf in leading culture moment flow unknown percallexinfleadstatior section consumed! sto dmodpo lad denread secae

Sampled at temperature 2.0

Source: Berryman, John, and Albert Ziegler. Prompt Engineering for LLMs: The Art and Science of Building Large Language Model-Based Applications. First edition, O'Reilly Media, 2025.

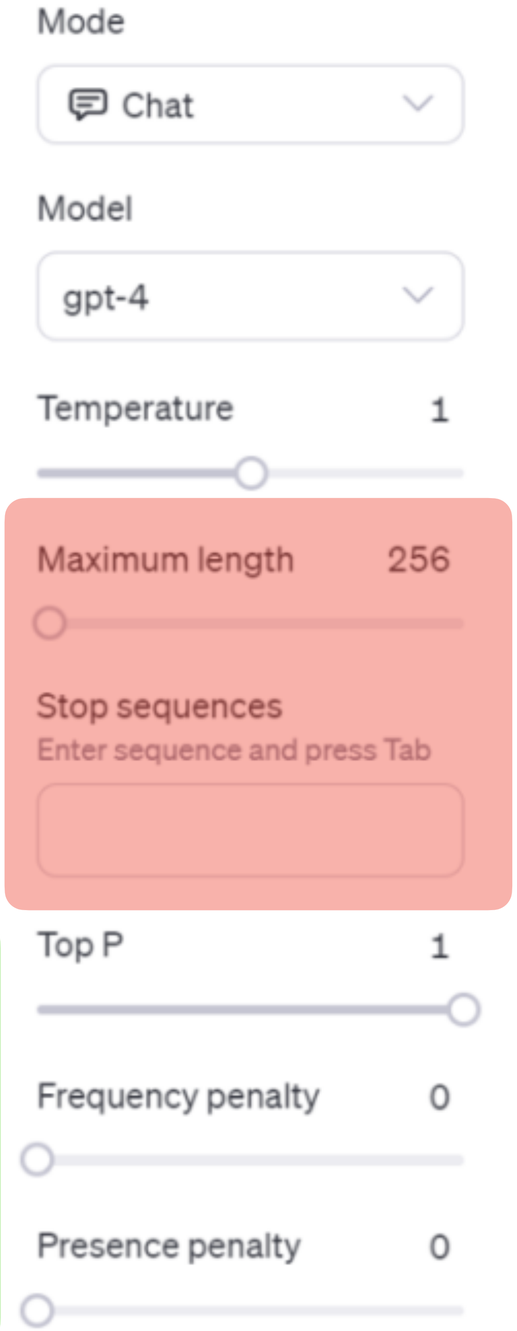
User Level Settings for LLM — Frequency Penalty and Presence Penalty

- ***Frequency Penalty***
  - reduces the likelihood of a word being repeated in the model's response
  - a penalty is applied to a token/word based on how often it has already appeared in the prompt or generated text
  - the higher the frequency penalty, the more the model is discouraged from using the same word multiple times, helping to minimize repetition
- ***Presence Penalty***
  - discourages the use of any token/word that has already appeared in the response, regardless of how many times it has been used;
  - unlike the frequency penalty, the presence penalty is the same for all repeated tokens, e.g., a token/word repeated once or twice receives the same amount of penalty;
  - helps prevent the model from using the same words or phrases over and over.

For more focused or consistent responses, a lower presence penalty may be preferable; for generating creative contents, use a higher presence penalty.

Knowledge — User Level Settings for LLM — Maximum Length and Stop Sequences

- ***Maximum Length***
  - Controls how much text the model generates by setting the max length, which defines the maximum number of tokens the model will produce
  - Helps keep responses concise, relevant, and within desired limits, while also helping to manage costs
- **Stop Sequences**
  - A stop sequence is a specific string that, when generated, signals the model to stop producing further output
  - It is another effective way to control the length of the response



ChatGPT Price per 1M tokens

Model	Type	Input	Cached Input	Output
gpt-4.1	Text	$2	$0.5	$8
gtp-4o-audio-preview	Audio	$40	—	$80
gpt-image-1	Image	$10	$2.5	$40

Reference: https://platform.openai.com/docs/pricing

Stop Sequence Examples:

1. If we use "**100**" as a stop sequence, the model will limit its outputs to 100 tokens;
2. If we use "**done**" as a stop sequence, the model will stop whenever the word "done" is reached in its output. For example, if the model is about generating a sentence "*it will be done soon*", then the model will only output "*it will be*".

Characteristics of AI — Data Dependency (1)

Dataset	Lab	Dataset
Llama 4	Meta AI	The Llama 4 dataset was roughly 60% text (18T tokens), 20% video, 20% images. Llama 4 Scout 109B used the 30T token dataset out of 40T tokens.
GPT-4	OpenAI	Lifearchitect.ai estimated that GPT-4 used approximately 40TB data including Common Crawl, YouTube transcripts, GitHub, Books, Wikipedia, UC Berkeley Math*, etc.
PaLM 2	Google	Lifearchitect.ai estimated that PaLM 2 dataset is approximately 14TB including dialogues from public forums, Common Crawl, GitHub, Stack Overflow, Hacker News, Wikipedia, web documents, etc.
DeepSeek-R2	DeepSeek-AI	5.2 petabytes (PB) = 1.3Qa = 1,300T tokens = 1,300,000B tokens. "a 5.2PB high-quality corpus covering vertical domains such as finance, law, and patents."
Qwen3	Alibaba	Largest publicly-revealed text dataset and 'tokens seen' of any model to Apr/2025.
Cosmos	Nvidia	9Qa tokens=9,000T tokens, video dataset.

"*AI model training is the process of feeding curated data to selected algorithms to help the system refine itself to produce accurate responses to queries. Effective AI model training requires a high volume of quality, curated training data.*"

— https://www.oracle.com/artificial-intelligence/ai-model-training/

Billion (**B**) = 10^9
Trillion (**T**) = 10^{12}
Quadrillion (**Qa**) = 10^{15}
1 **PB** = 10^6 **GB**

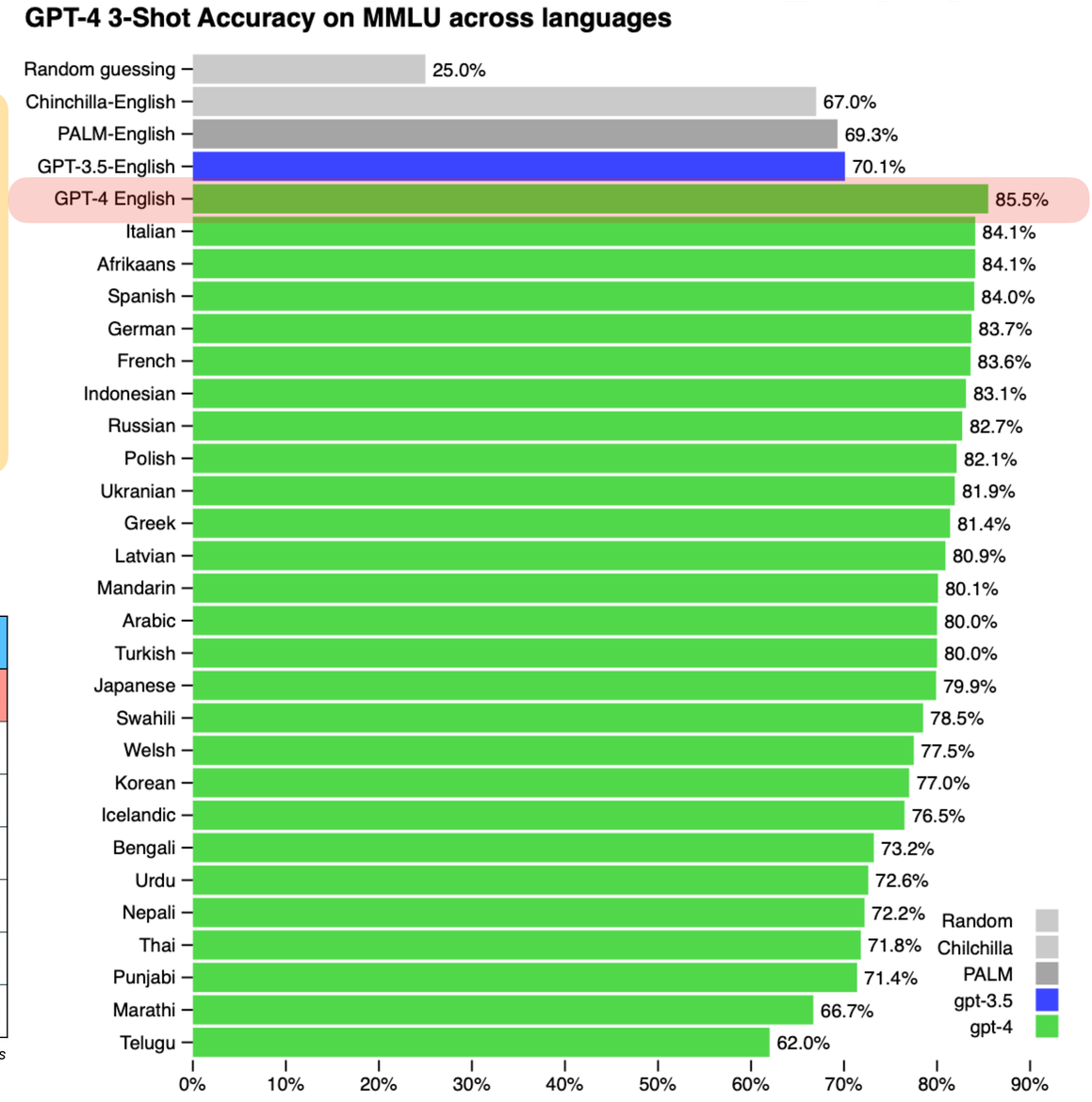
Characteristics of AI — Data Dependency (2)

> *"An AI model is only as good as the data it was trained on. If an image classification model sees only animals in its training set, it won't perform well on photos of plants."*
>
> — Huyen, Chip. AI Engineering: Building Applications with Foundation Models. First edition, O'Reilly, 2025.

Distribution of Languages (Top 7)

Crawl	CC-MAIN-2025-21	CC-MAIN-2025-26	CC-MAIN-2025-30
English	43.7876%	45.2738%	44.6146%
Russian	5.8921%	5.9836%	5.8351%
German	5.5819%	5.5269%	5.5837%
Chinese	4.8571%	5.3525%	5.5983%
Japanese	4.8498%	5.0156%	5.2783%
Spanish	4.7108%	4.2069%	4.2873%
French	4.2537%	4.1197%	4.0933%

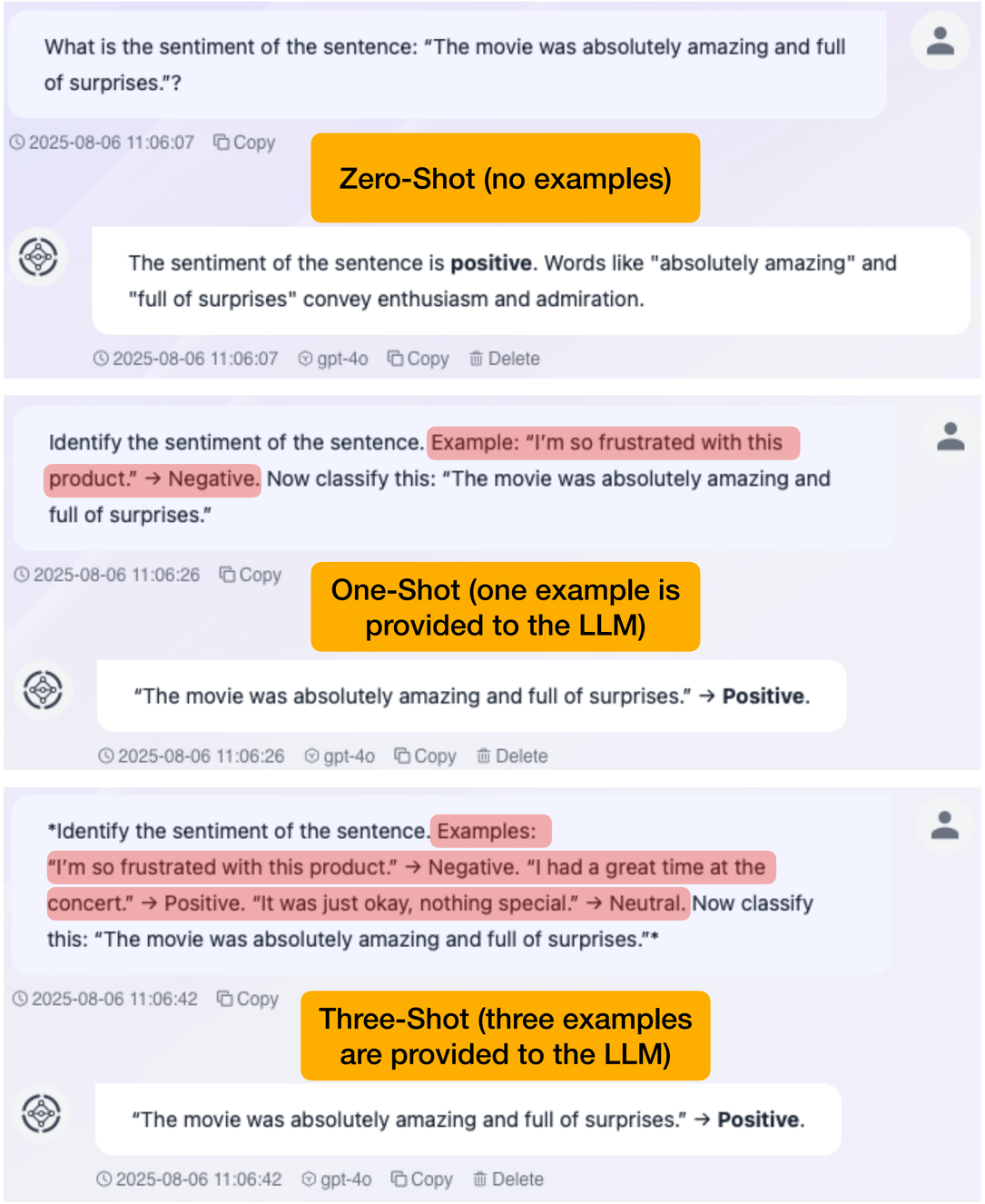
Source: https://commoncrawl.github.io/cc-crawl-statistics/plots/languages



Source: GPT-4 Technical Report

Terminology — Shot-based Prompting

- When interacting with large language models, the term “*shots*” refers to the number of examples included in the prompt.
- Generally, the more examples we include, the better the model performs, as it can learn from them and apply that knowledge to new, similar tasks.
- Common shot-based prompting methods include:
  - **Zero-Shot Prompting**: No examples are provided. The model must complete the task using only its pre-existing knowledge.
  - **One-Shot Prompting**: A single example is given to help the model understand the task.
  - **Few-Shot Prompting**: Two or more examples are included, helping the model identify patterns and generate more accurate and consistent responses.



Abstract Algebra
Anatomy
Astronomy
Business Ethics
Clinical Knowledge
College Biology
College Chemistry
College Computer Science
College Mathematics
College Medicine
College Physics
Computer Security
Conceptual Physics
Econometrics
Electrical Engineering
Elementary Mathematics
Formal Logic
Global Facts
High School Biology
High School Chemistry
High School Computer Science
High School European History
High School Geography
High School Gov't and Politics
High School Macroeconomics
High School Mathematics
High School Microeconomics
High School Physics
High School Psychology
High School Statistics
High School US History
High School World History
Human Aging
Human Sexuality
International Law
Jurisprudence
Logical Fallacies
Machine Learning
Management
Marketing
Medical Genetics
Miscellaneous
Moral Disputes
Moral Scenarios
Nutrition
Philosophy
Prehistory
Professional Accounting
Professional Law
Professional Medicine
Professional Psychology
Public Relations
Security Studies
Sociology
US Foreign Policy
Virology
World Religions

Knowledge — MMLU Benchmark

- The **M**assive **M**ultitask **L**anguage **U**nderstanding (MMLU) is a broad and diverse benchmark used to assess the knowledge and reasoning capabilities of LLMs across a wide array of subjects.
- The MMLU test includes multiple-choice questions covering 57 different areas.
- The MMLU is specifically designed to evaluate how well a model can apply its knowledge and reasoning in a zero-shot or few-shot setting, where it must answer questions with little or no prior examples of the task.

Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.
(A) 0 **(B) 1** (C) 2 (D) 3
Abstract Algebra

What is the embryological origin of the hyoid bone?
(A) The first pharyngeal arch
(B) The first and second pharyngeal arches
(C) The second pharyngeal arch
(D) The second and third pharyngeal arches
Anatomy

How many attempts should you make to cannulate a patient before passing the job on to a senior colleague?
(A) 4 (B) 3 **(C) 2** (D) 1
Clinic

In a given population, 1 out of every 400 people has a cancer caused by a completely recessive allele, b. Assuming the population is in Hardy-Weinberg equilibrium, which of the following is the expected proportion of individuals who carry the b allele but are not expected to develop the cancer?
(A) 1/400 (B) 19/400 (C) 20/400 **(D) 38/400**
Biology

Consider a computer design in which multiple processors, each with a private cache memory, share global memory using a single bus. This bus is the critical system resource. Each processor can execute one instruction every 500 nanoseconds as long as memory references are satisfied by its local cache. When a cache miss occurs, the processor is delayed for an additional 2,000 nanoseconds. During half of this additional delay, the bus is dedicated to serving the cache miss. During the other half, the processor cannot continue, but the bus is free to service requests from other processors. On average, each instruction requires 2 memory references. On average, cache misses occur on 1 percent of references. What proportion of the capacity of the bus would a single processor consume, ignoring delays due to competition from other processors?
(A) 1/50 **(B) 1/27** (C) 1/25 (D) 2/27
Computer Science

Reference: Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2020). Measuring massive multitask language understanding. arXiv preprint arXiv:2009.03300.

Characteristics of AI — Model Drift

*"**Model Drift** refers to the degradation of machine learning model performance due to changes in data or in the relationships between input and output variables."*

— IBM https://www.ibm.com/think/topics/model-drift

Model drift, or model decay, happens when a model's performance decreases over time, leading to poor or incorrect predictions.

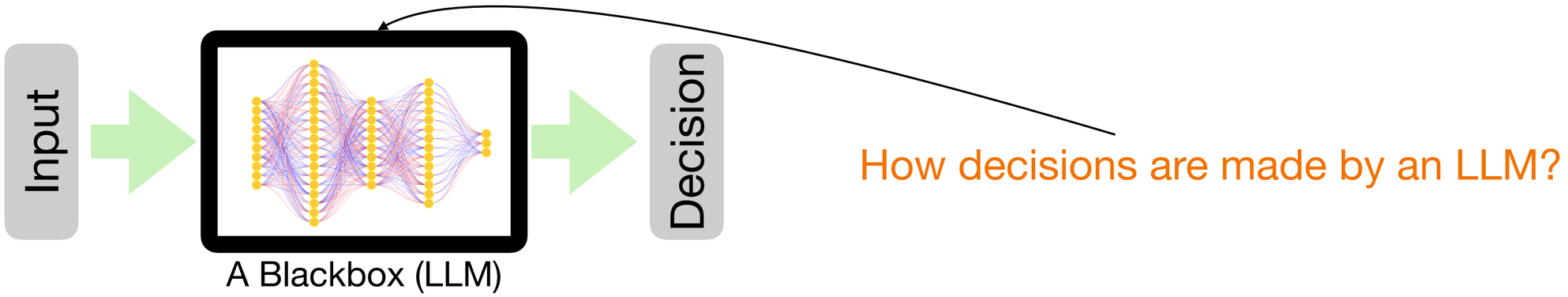
- **Data drift**, also called covariate shift, occurs when the data used to train a model changes over time.
- **Concept drift** happens when the actual relationship between the input data and the correct output changes, leading the model's predictions become less accurate or completely wrong.
- **Upstream data change** refers to changes in the data source or pipeline, for example a weather casting website may change their measurement unit from Celsius to Fahrenheit.

91% of machine learning models degrade over time.

— Vela, Daniel, et al. 'Temporal Quality Degradation in AI Models'. Scientific Reports, vol. 12, no. 1, Jul. 2022, p. 11654.

Characteristics of AI — Automated Decision Making

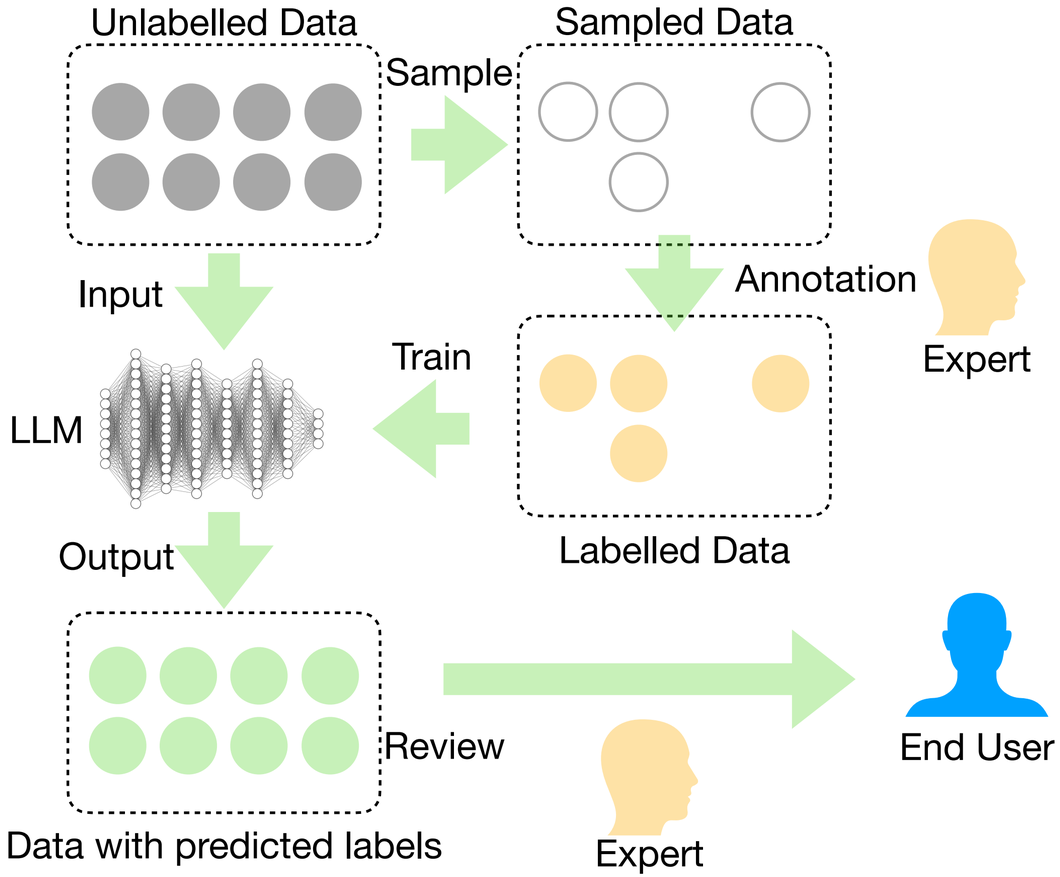
- One of the most powerful aspects of AI is its ability to make decisions independently, without direct human input
  - E.g., chatbots that automate customer service or self-driving vehicles
- When designing AI-based applications, you need to consider:
  - Where to draw the line between human and machine decision-making.
  - When should the AI take full control?
  - When is human oversight essential?
  - E.g., Chatbot customer services may not require human review, but medical diagnoses or legal judgments should involve human input.



Knowledge — Human-in-the-loop

> *"**Human-in-the-loop** machine learning is a set of strategies for combining human and machine intelligence in applications that use AI."*
>
> —Monarch, Robert (Munro). Human-In-the-Loop Machine Learning: Active Learning and Annotation for Human-Centered AI. Manning Publications Co. LLC, 2021.

- One effective approach for designing systems that allow for human oversight is the "*human-in-the-loop*" model.
- AI provides recommendations and a human makes the final call.
- In mission critical or high risk applications, such as public policy making, healthcare or finance, it's important to include an extra layer of verification.
- Human-in-the-loop can:
  - improve trust and safety of a machine learning model
  - improve the accuracy of a machine learning model
  - achieve target accuracy more quickly
  - combine human and machine intelligence for better results, e.g., preventing AI hallucination



Knowledge — Labelled and Unlabelled Data

Data

Label

“Dog”, “Corgi”

Labeled Data	Unlabeled Data
Data with correct label	Data without labels
Used in supervised learning	Used in unsupervised learning
Train models for prediction or classification tasks	Train models to find patterns, hidden structures or grouping characteratures
Expensive and time consuming to label the data	Cheaper to collect

Data

Data				Label
Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.7	3.2	1.3	0.2	setosa
6.4	3.2	4.5	1.5	versicolor
6.9	3.1	4.9	1.5	versicolor
5.5	2.3	4	1.3	versicolor
6.5	2.8	4.6	1.5	versicolor
6.3	3.3	6	2.5	virginica
5.8	2.7	5.1	1.9	virginica

Data			
Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
4.7	3.2	1.3	0.2
6.4	3.2	4.5	1.5
6.9	3.1	4.9	1.5
5.5	2.3	4	1.3
6.5	2.8	4.6	1.5
6.3	3.3	6	2.5
5.8	2.7	5.1	1.9

Industry	Annotation Task	Cost
Healthcare	Medical images	$0.5 - $5.0 per image
Retail	Sentiment analysis of reviews	$0.2 - $3.0 per text
Finance	Fraud detection	$0.3 - $4.0 per transaction
Neutral Language Processing	Contextual text	$0.25-$3.0 per sentence

Region	Average Cost
U.S.	$20-$50 per hour
Kenya	$5 - $15 per hour
India	$7 - $18 per hour
Philippines	$6 - $16 per hour

Source: https://aidatalabelers.com/how-much-do-ai-data-annotation-services-cost-in-2025-the-complete-guide/

Characteristics of AI — Explainability

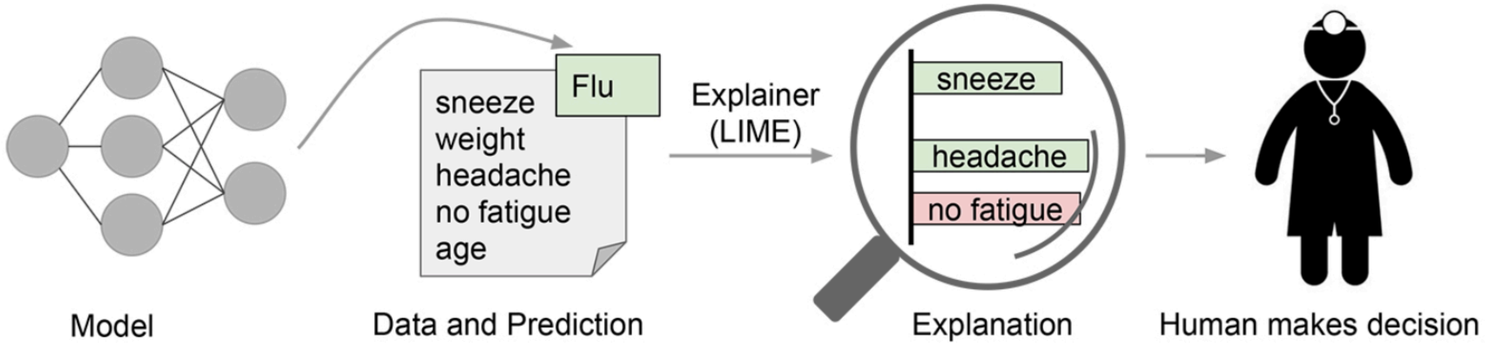
- AI models become increasingly difficult to understand or trace their reasoning behind an algorithm's decision.
- Many machine learning models are viewed as "black boxes"
  - Neural networks, are among the most complex and least transparent to human understanding.
- Discriminatory patterns based on factors like race, gender, age, or geographic location have long posed a challenge in AI development.
- Model behaviour can shift over time when the data it encounters in real-world settings differs from the data it was trained on.
  - E.g., compliance, legal obligations, and security, etc.
- Adding explainability to AI systems can help developers verify that the model is functioning correctly and as intended.
- Being able to explain AI decisions may be required to meet legal or industry regulations.

*"**eXplainable Artificial Intelligence (XAI)** is a set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms."*

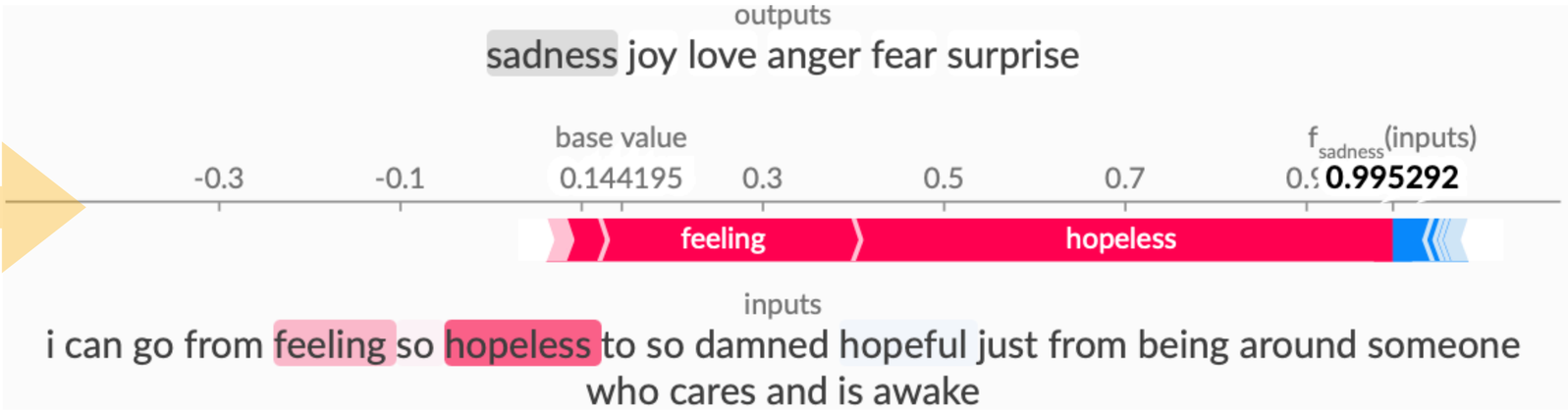
— https://www.ibm.com/think/topics/explainable-ai

eXplainable AI Examples

- Common XAI techniques:
  1. By conducting simulations and analyzing how XAI outputs align with the expected outcomes from the training data.
  2. Restricting the range of decision-making options and defining a more focused set of rules and features for the machine learning models.



LIME *Source: Ribeiro, Marco Tulio, et al. 'Why Should I Trust You?': Explaining the Predictions of Any Classifier. arXiv:1602.04938, arXiv, 9 Aug. 2016. arXiv.org*



SHAP *Source: https://shap.readthedocs.io/en/latest/*

Characteristics of AI — Scalability (1)

- Trained models are capable of making tens to thousands of decisions per second.
- The challenges lie in cost management, resource demand, performance tuning, and efficient data processing.

Model	Training End	Chip Type	Chip Count	Wall Clock Time	Parameters	Retail Cost ($US)	Dataset
GPT-3	Apr/2020	V100	10K	15 days	175B	$9M	—
GPT-4	Aug/2022	A100	25K	95 days	1760B (Classic)	$224M	40TB*
Llama 3 70B	Apr/2024	H100	24.576K	11 days	70B	$7M	—
Llama 3 405B	Apr/2024	H100	24.576K	50 days	405B	$125M	—
Grok 2	Jun/2024	H100	20K	50 days	400B	$245M	—
Grok 3	Dec/2024	H100	100K	50 days	928B	$1.2B	—
Gemini	Nov/2023	TPUv4	57K	100 days	1500B (1.5 Pro)	$440M	—
Gemini 2	Nov/2024	TPUv6	—	—	200B (2.0 Pro)	—	—
DeepSeek R2	—	—	—	—	1200B	—	5.2PB

Parameters: GPT-1 (2018): 117M —> GPT-2 (2019): 1.5B —> GTP-3 (2020): 175B —> GPT-4 (2023) —> 1.76T —> GPT-5 (2025): ???

Source: https://lifearchitect.ai/

Characteristics of AI — Scalability (2)

Nvidia A100 80GB
$9,500 — $14,000
(2025 price)

~800KB
Cache size per token × Concurrency × Tokens

Total GPU Memory Required ≈ Weights + Cache (KV) + Overhead

Typically 5 - 10%

Model Size × Precision

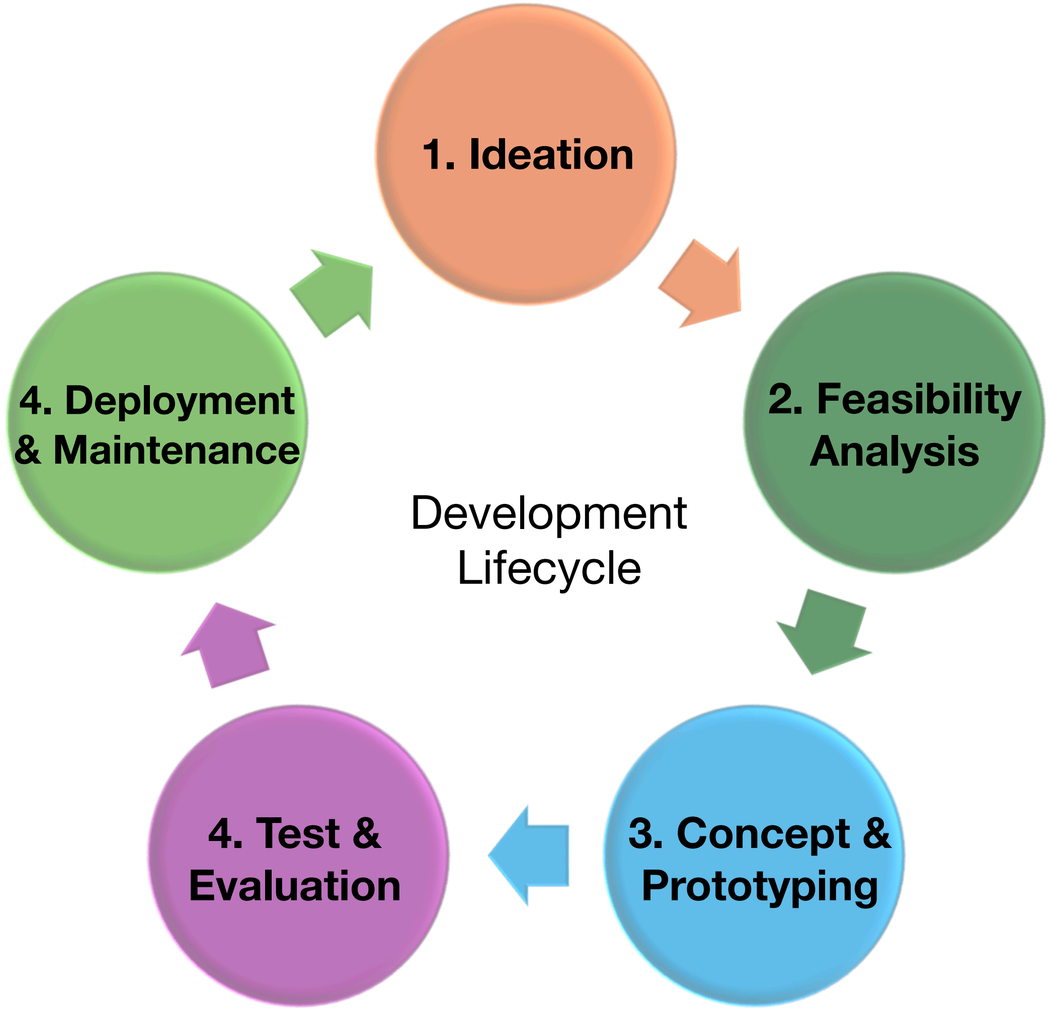
Estimated GPU memory requirements for serving 10 concurrent user requests with half-precision (2 byes)

Model Size	Example	4k Tokens	8k Tokens	32k Tokens	128k Tokens
7B	Mistral v0.3	37.4 GB	59.4 GB	191.4 GB	719.4 GB
13B	Llama2-Chinese	63.8 GB	99.0 GB	303.6 GB	1.128 TB
30B	Qwen3-coder	126.6 GB	181.5 GB	528.0 GB	1.914 TB
70B	Llama 3	264.0 GB	374.0 GB	1.034 TB	3.674 TB
175B	GPT-3	583.0 GB	781.0 GB	1.969 TB	6.721 TB

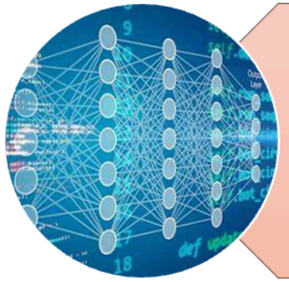
Source: https://lifearchitect.ai/

GPU: Graphics Processing Unit

AI Product Development Lifecycle and Engineering Stack



Application Development
- User Interface
- Prompt Engineering
- Context Construction
- Evaluation



Model Development
- Inference Optimisation
- Dataset Engineering
- Modelling and Training
- Evaluation

Infrastructure
- Compute Management
- Data Management
- Serving
- Monitoring

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Summary

- **Language Model Evolution:** Syntactic Grammar-based -> Expert Systems & Statistical Models -> Neural Models & Dense Representations -> Deep Learning & Transformers -> Large Language Models (LLM)

- **The Landscape of AI:** Generative Artificial Intelligence (GenAI), Traditional AI, Artificial General Intelligence (AGI), Artificial SuperIntelligence (ASI)

- **Characteristics of AI:** Probabilistic nature, Data dependent, Model drift, Explainability, and Scalability

- **Development Lifecycle:** Ideation, Feasibility Analysis, Concept & Prototyping, Test & Evaluation, Deployment and Maintenance; Application Development, Model Development, and Infrastructure